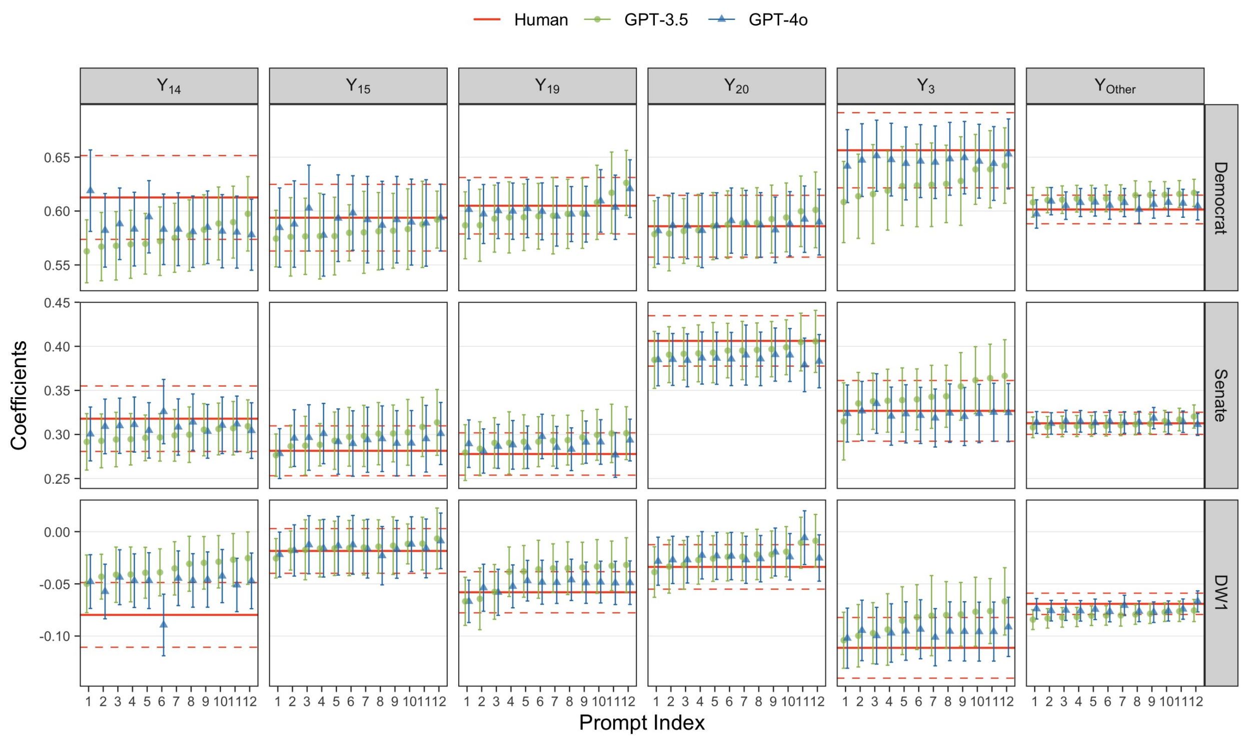
# Results RHS

Aug 12, 2024

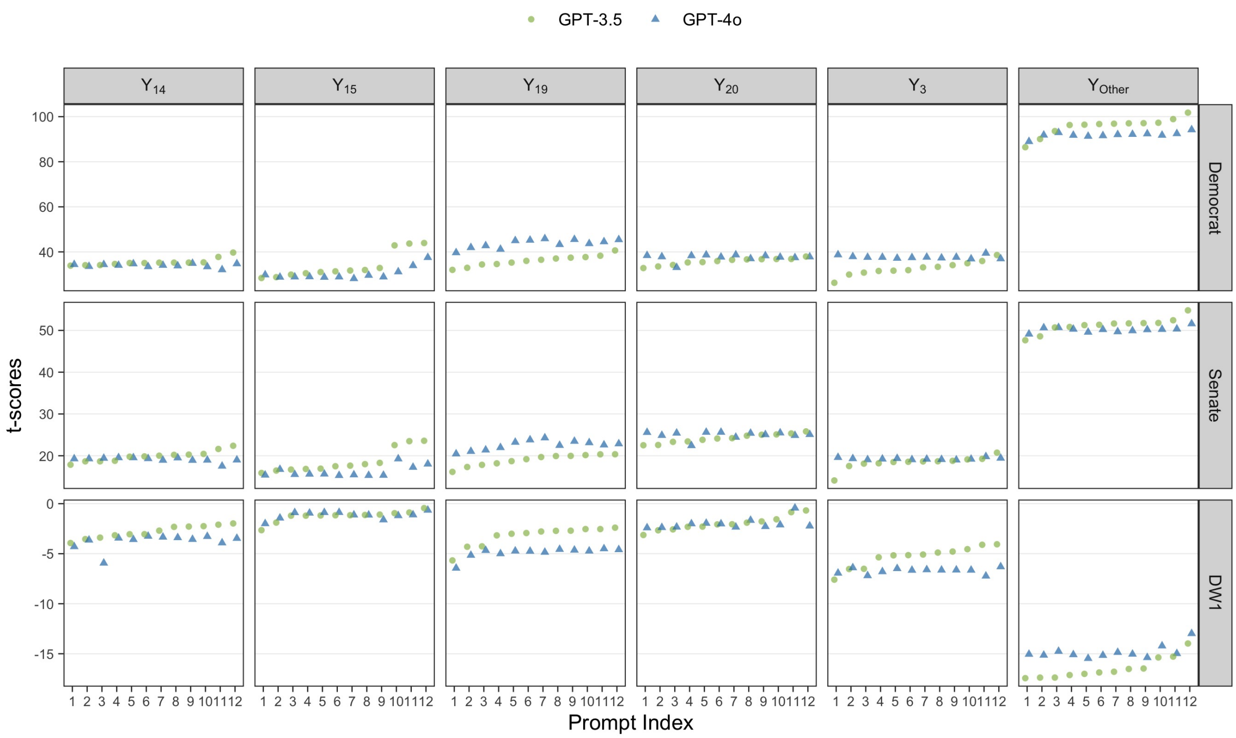
## Proxy on the RHS

Figure 1: $\beta_t$ vs. Prompt for all $\beta$s. Proxy on the RHS.



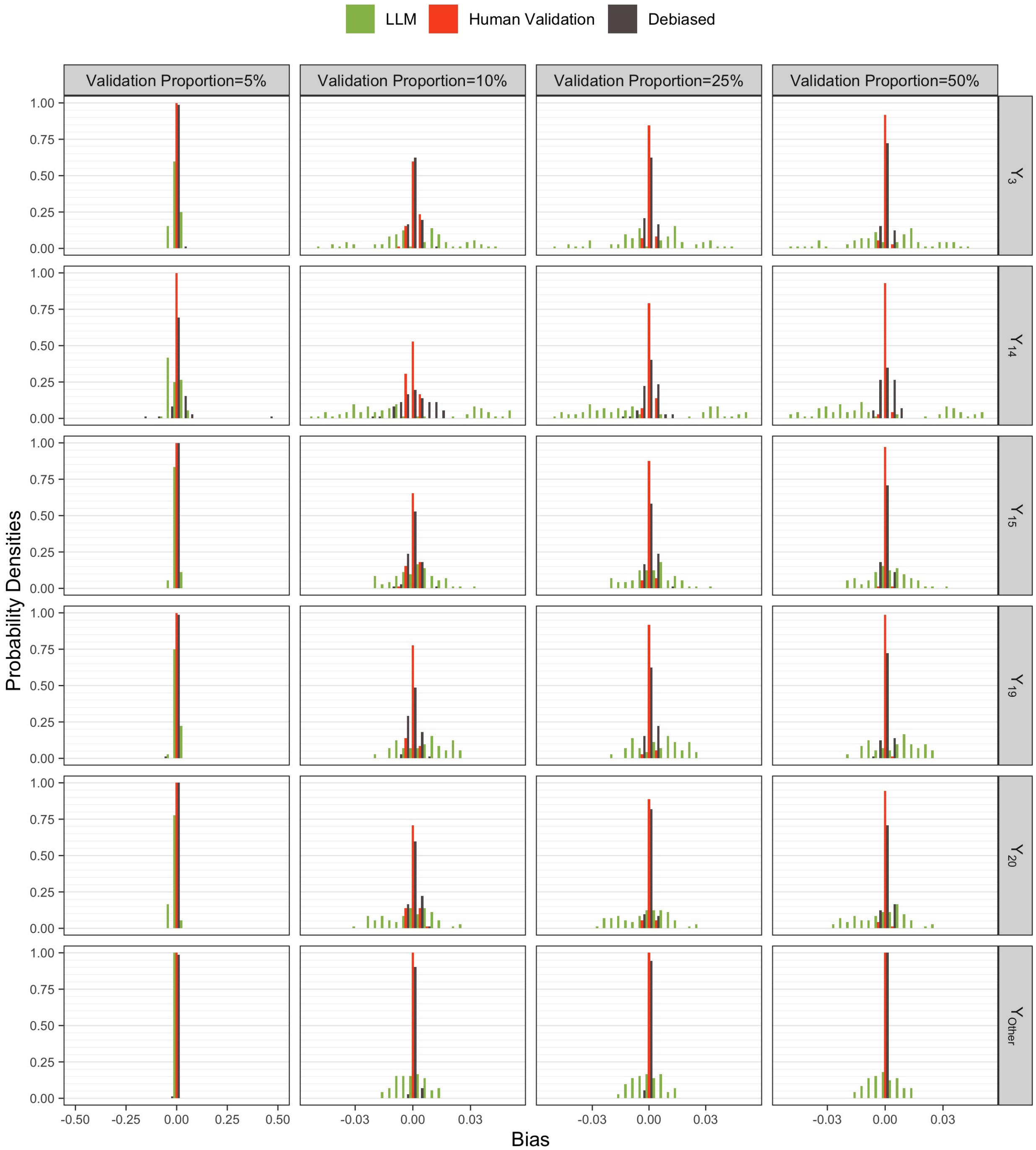

Note: The coefficients were obtained by fitting the model $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, $Y_t$ denotes the indicator for topic $t$, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable $V$ takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, Senate $= 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 2: t-scores of each $\beta_t$ vs. Prompt. Proxy on the RHS.



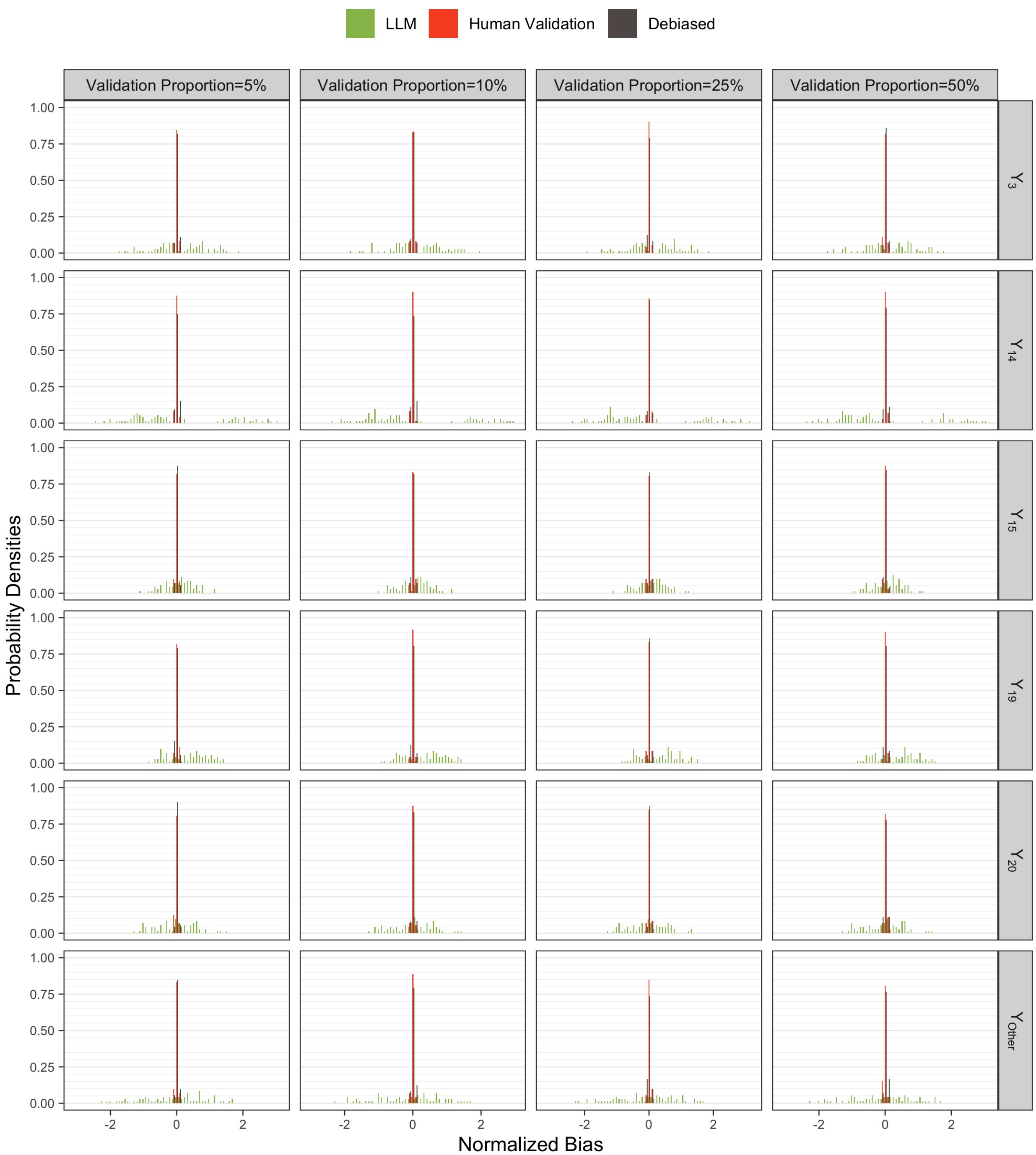

Note: The t-scores were obtained by fitting the model $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, $Y_t$ denotes the indicator for topic $t$, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable $V$ takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 3: Bias Density Plots for all $\beta$s by Proportion of Validation Samples. Proxy on the RHS.




Note: The bias estimates were computed as the difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{V,\eta,m,p} = \sum_{i=1}^{N} \beta^{(i)}_{t,V,\eta,m,p}/N$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, using a regression model of the form $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$, for topic $t \in \{3, 14, 15, 19, 20, \text{Other}\}$, variable $V \in \{\text{Democrat}, \text{Senate}, \text{DW1}\}$, validation proportion $\eta \in \{5\%, 10\%, 25\%, 50\%\}$, LLM model $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$, and prompt $p \in \{1, \ldots, 12\}$. For the LLM-based models (green), a random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels, $Y^{\text{LLM}}_{t,m,p} = 1\{Y^{\text{LLM}}_{m,p} = t\}$. For the other two models—human-based (red) and debiased-based (brown)—the 5,000 sample were split into two subsets: validation and primary, with the size of the validation set determined by the validation proportion, $\eta$. In the human-based model, only the human labels in the validation set were used, $Y^{\text{Human}}_t = 1\{Y^{\text{Human}} = t\}$, whereas the debiased estimates use both the validation and primary sets, with $Y^{\text{Human}}_t$ from validation set used to estimate measurement errors and $Y^{\text{LLM}}_{t,m,p}$ from the primary set used to obtain the coefficient estimates used to compute the reported bias.

Figure 4: Normalized Bias Density of $\beta_1$ by Proportion of Validation Samples. Proxy on the LHS.




Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from $N =$ 1000 simulation runs, $\bar{\beta}_{t,V,\eta,m,p}$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, divided by the sample standard deviation, $s_{t,V,\eta,m,p}$, where $s^2_{t,V,\eta,m,p} = \sum_{i=1}^{N}(\beta^{(i)}_{t,V,\eta,m,p} - \bar{\beta}_{t,V,\eta,m,p})^2/(N-1)$ and all variables are as defined previously.

Figure 5: Bias Density of $\beta_3$ by Model and Proportion of Validation Samples. Proxy on the RHS.

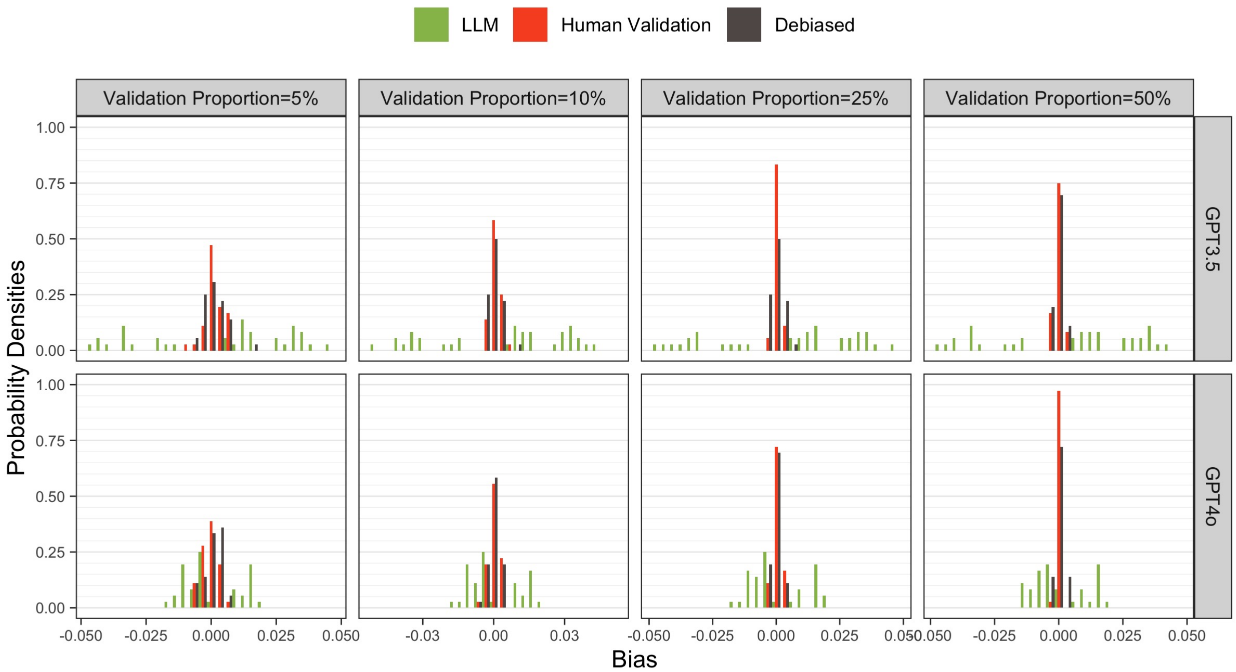


Figure 6: Bias Density of $\beta_{14}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

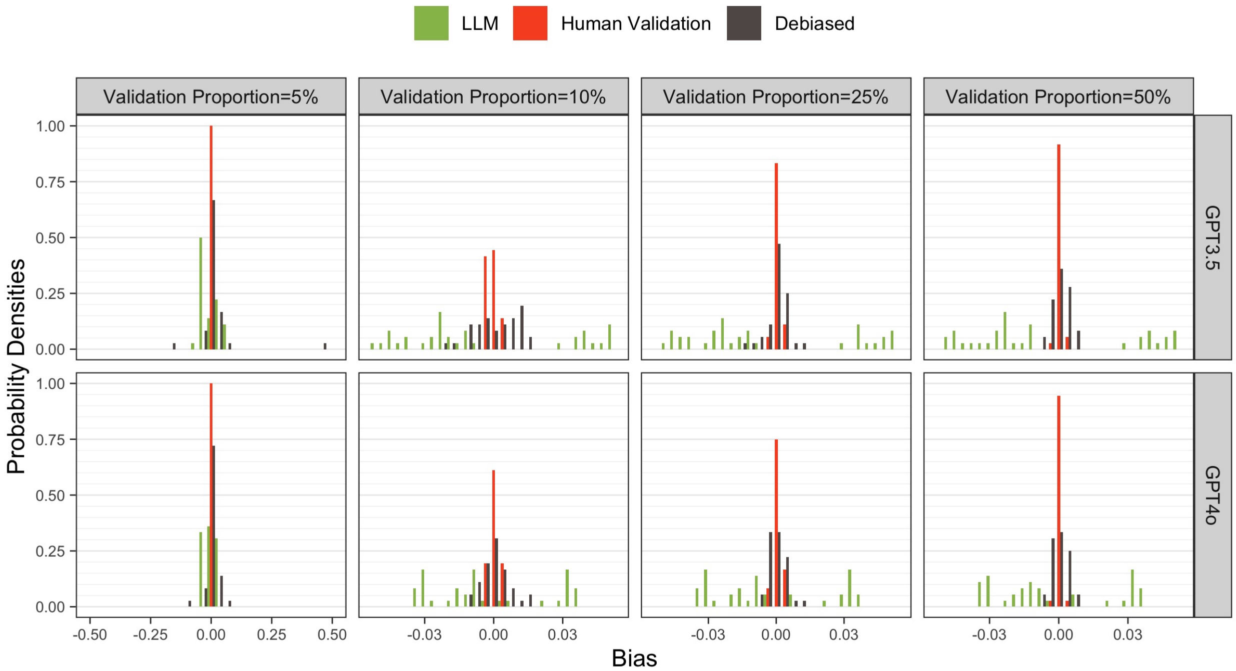



Figure 7: Bias Density of $\beta_{15}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

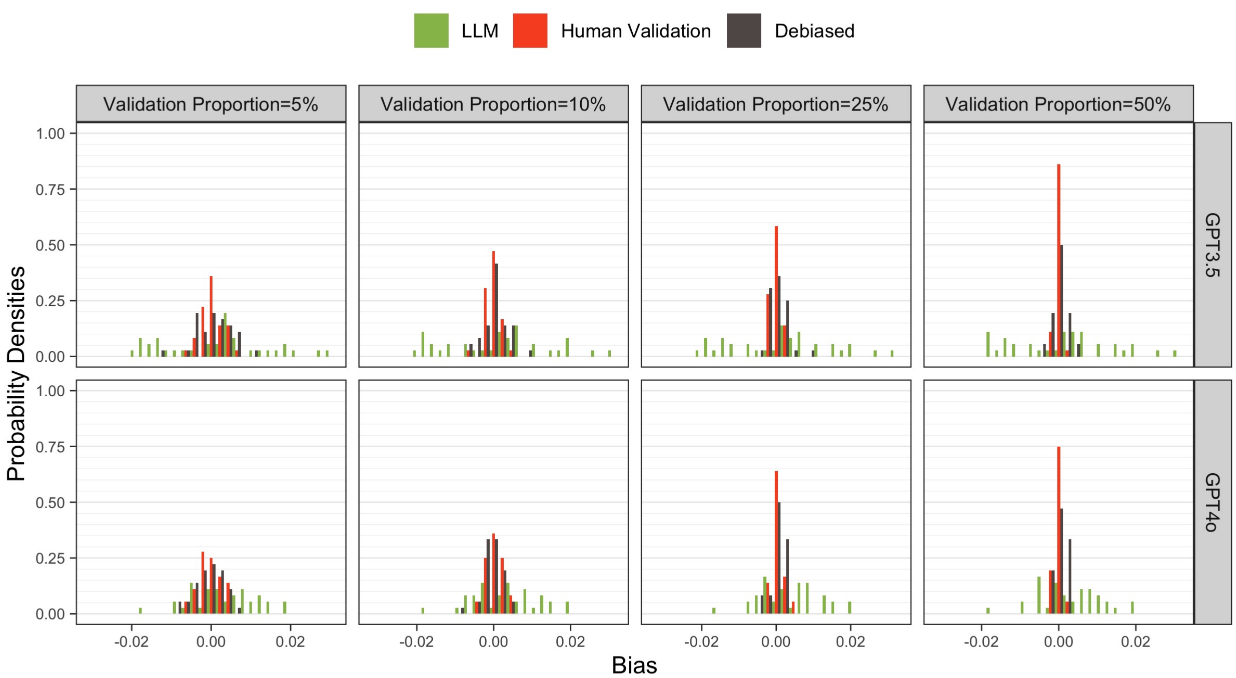


Figure 8: Bias Density of $\beta_{19}$ by Model and Proportion of Validation Samples. Proxy on the RHS.


Figure 9: Bias Density of $\beta_{20}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

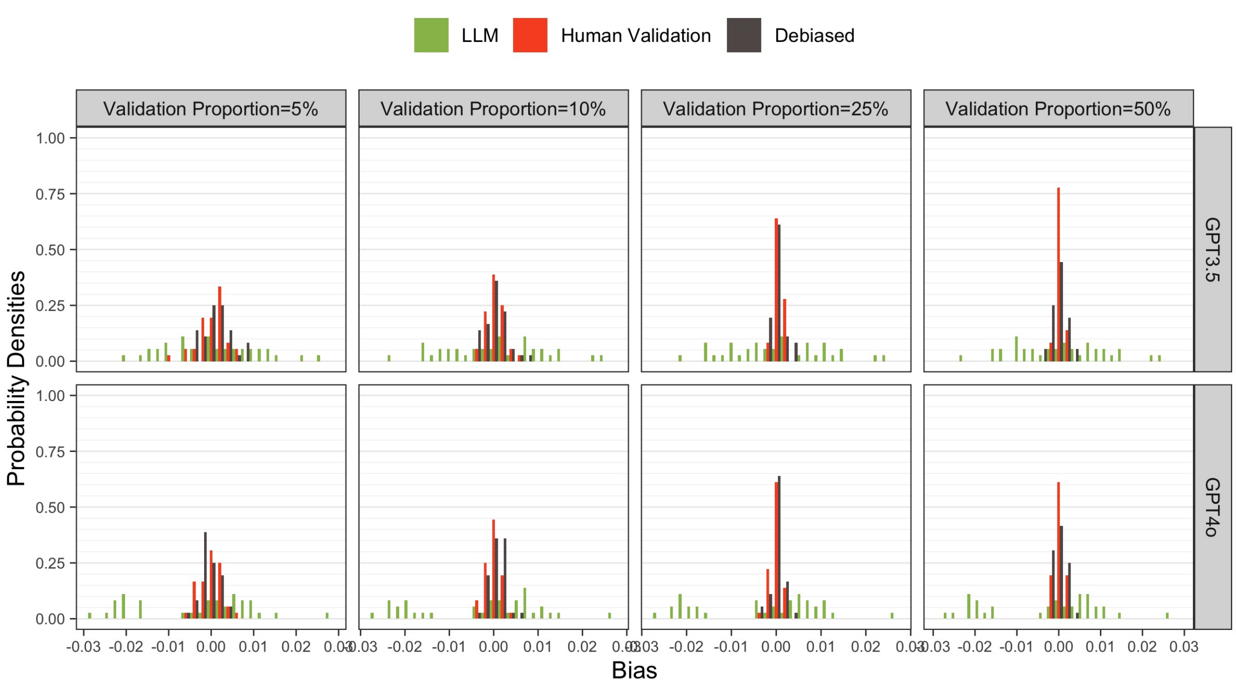


Figure 10: Bias Density of $\beta_{\text{Other}}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

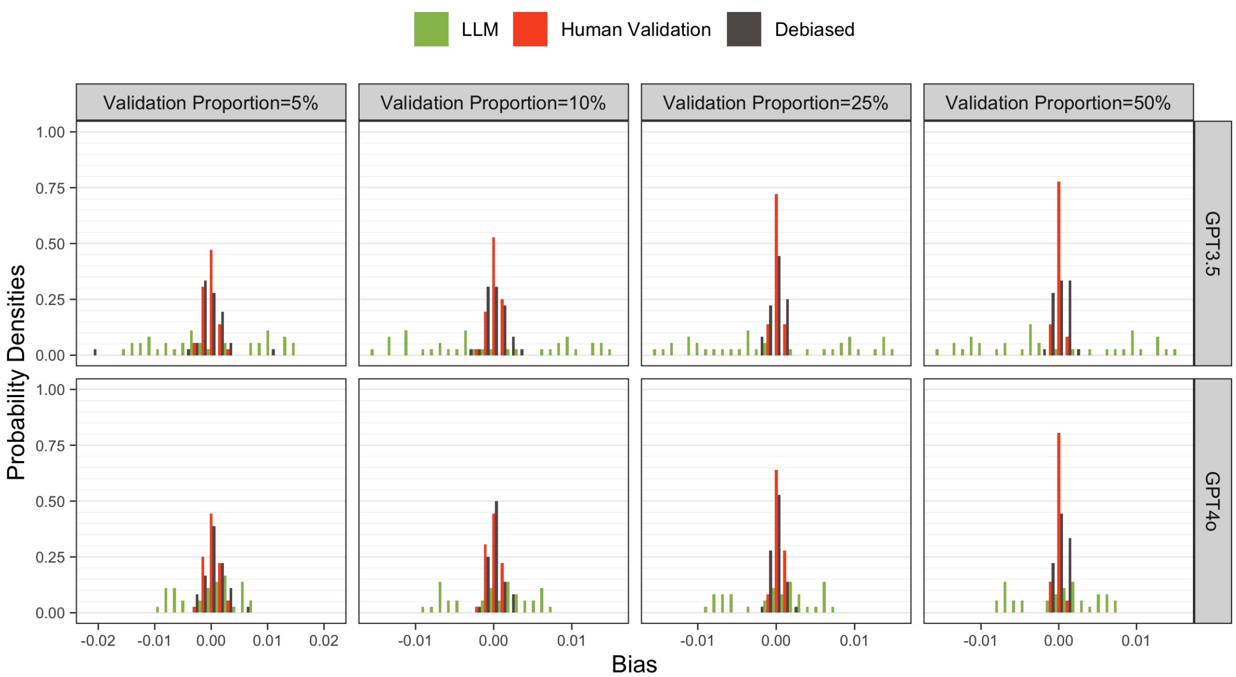


Figure 11: Normalized Bias Density of $\beta_3$ by Model and Proportion of Validation Samples. Proxy on the RHS.

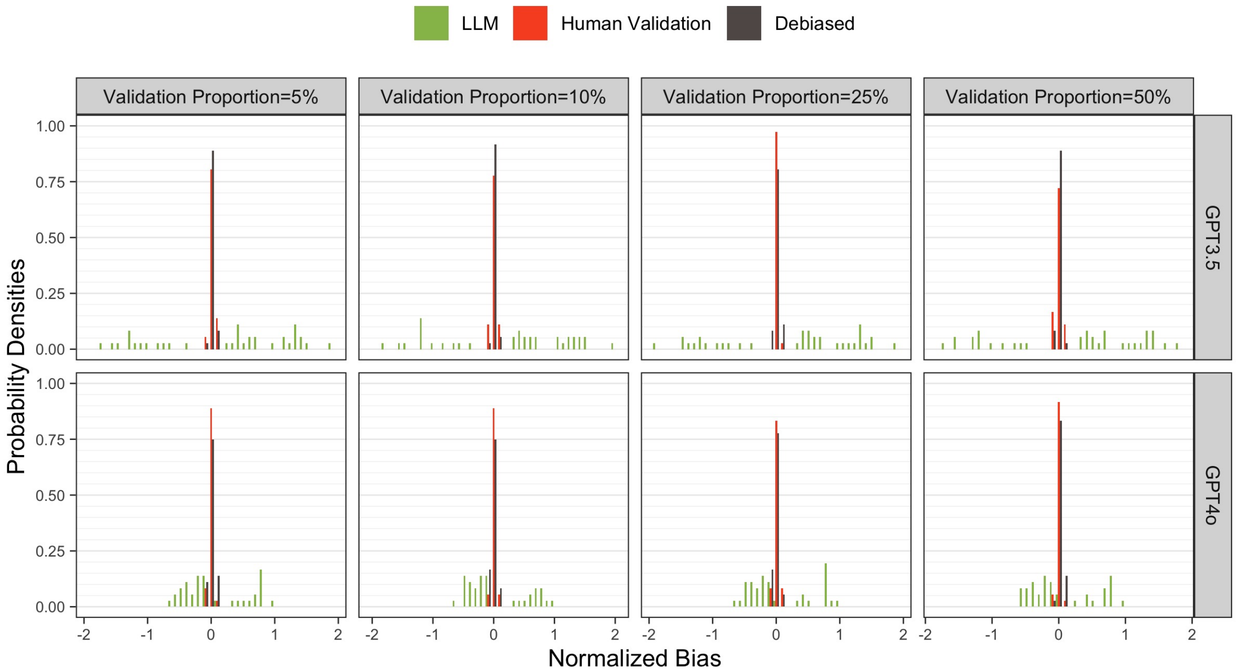


Figure 12: Normalized Bias Density of $\beta_{14}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

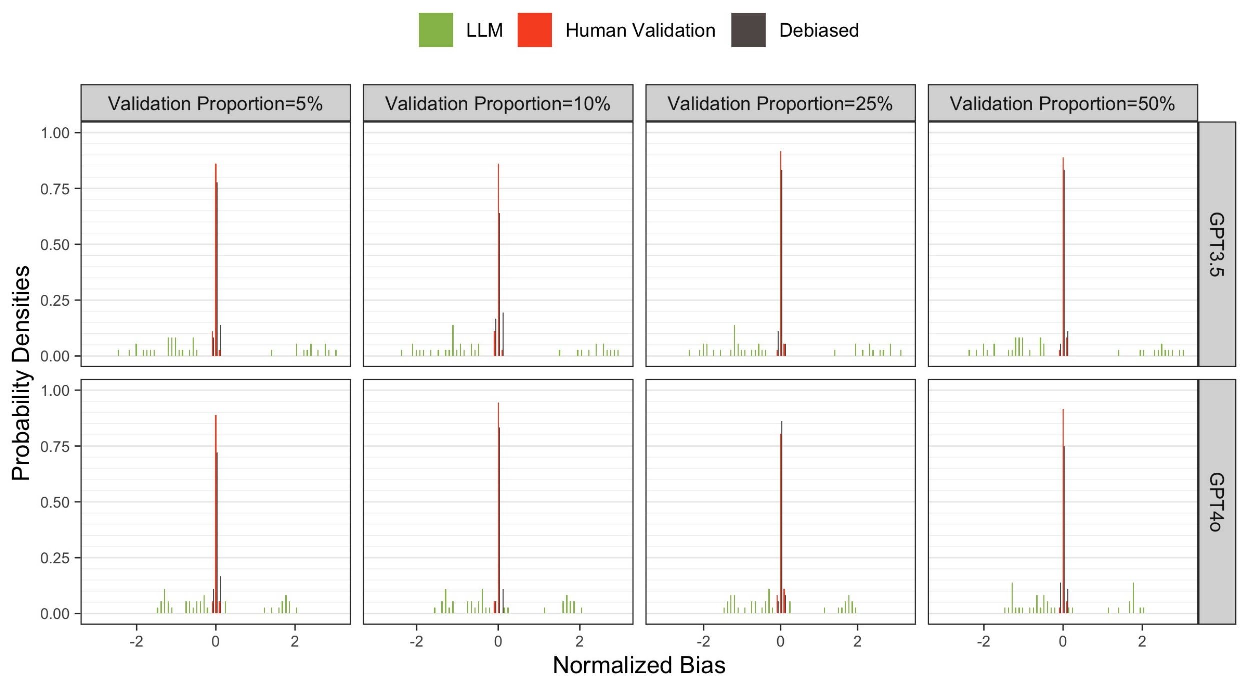



Figure 13: Normalized Bias Density of $\beta_{15}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

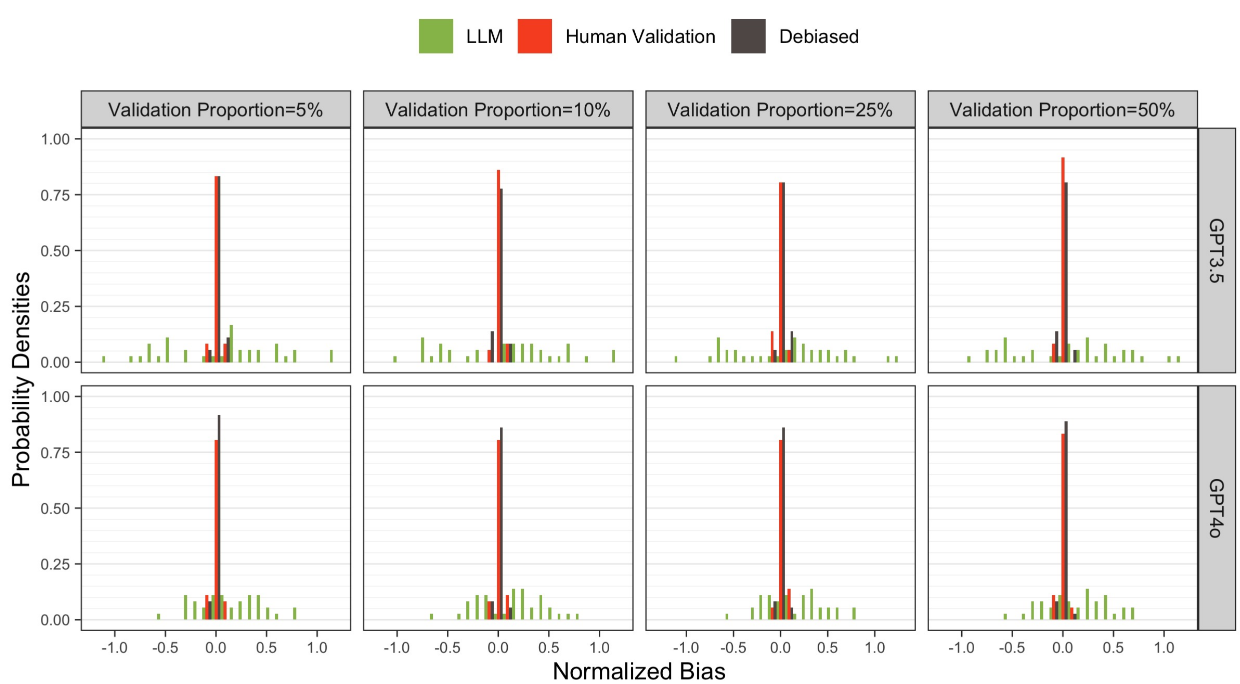


Figure 14: Normalized Bias Density of $\beta_{19}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

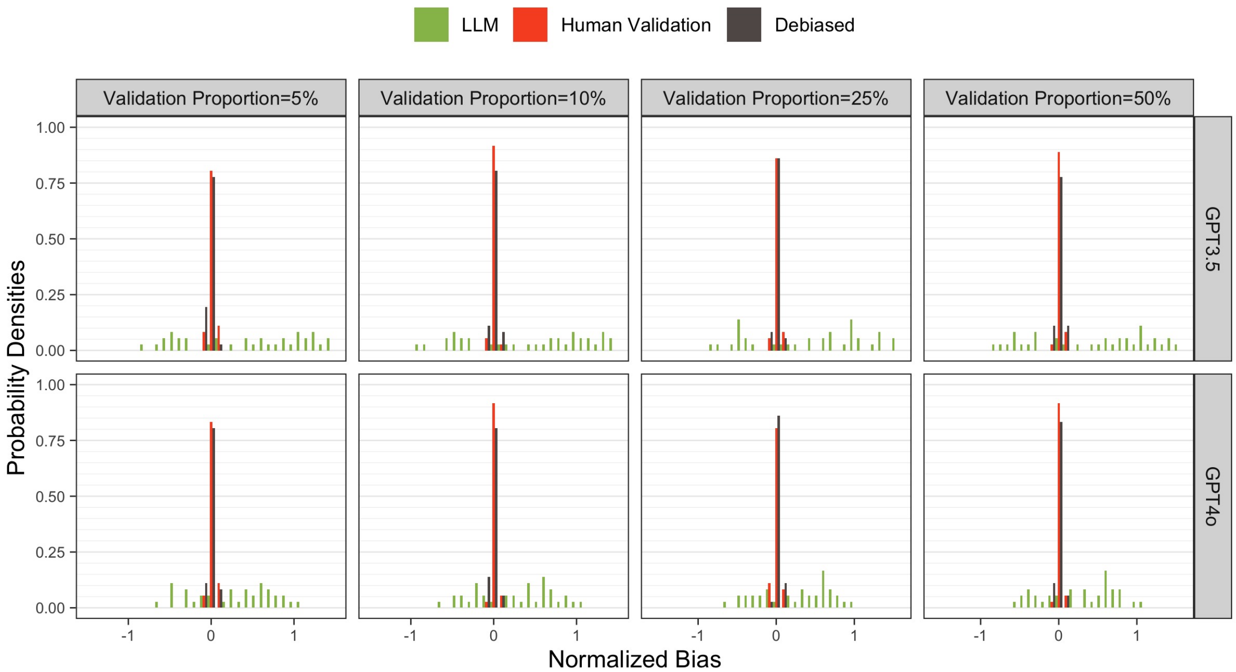



Figure 15: Normalized Bias Density of $\beta_{20}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

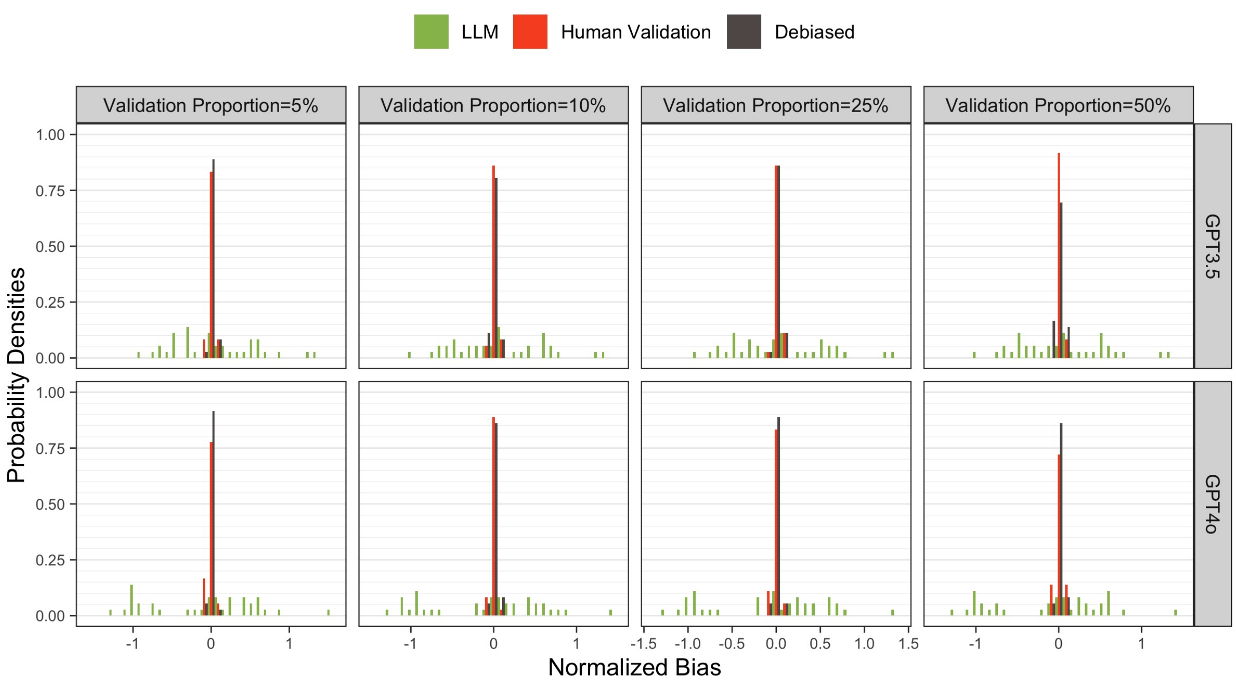


Figure 16: Normalized Bias Density of $\beta_{\text{Other}}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

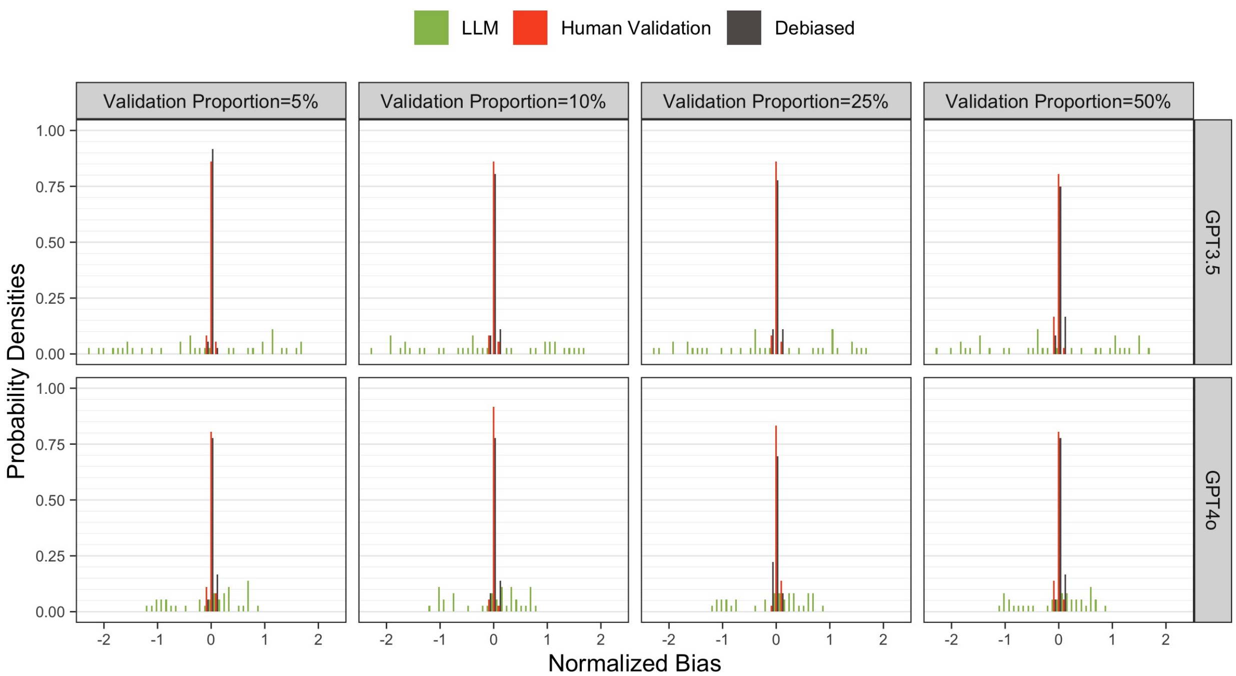


Figure 17: MSE Empirical CDF for all $\beta$s by Proportion of Validation Samples. Proxy on the RHS.

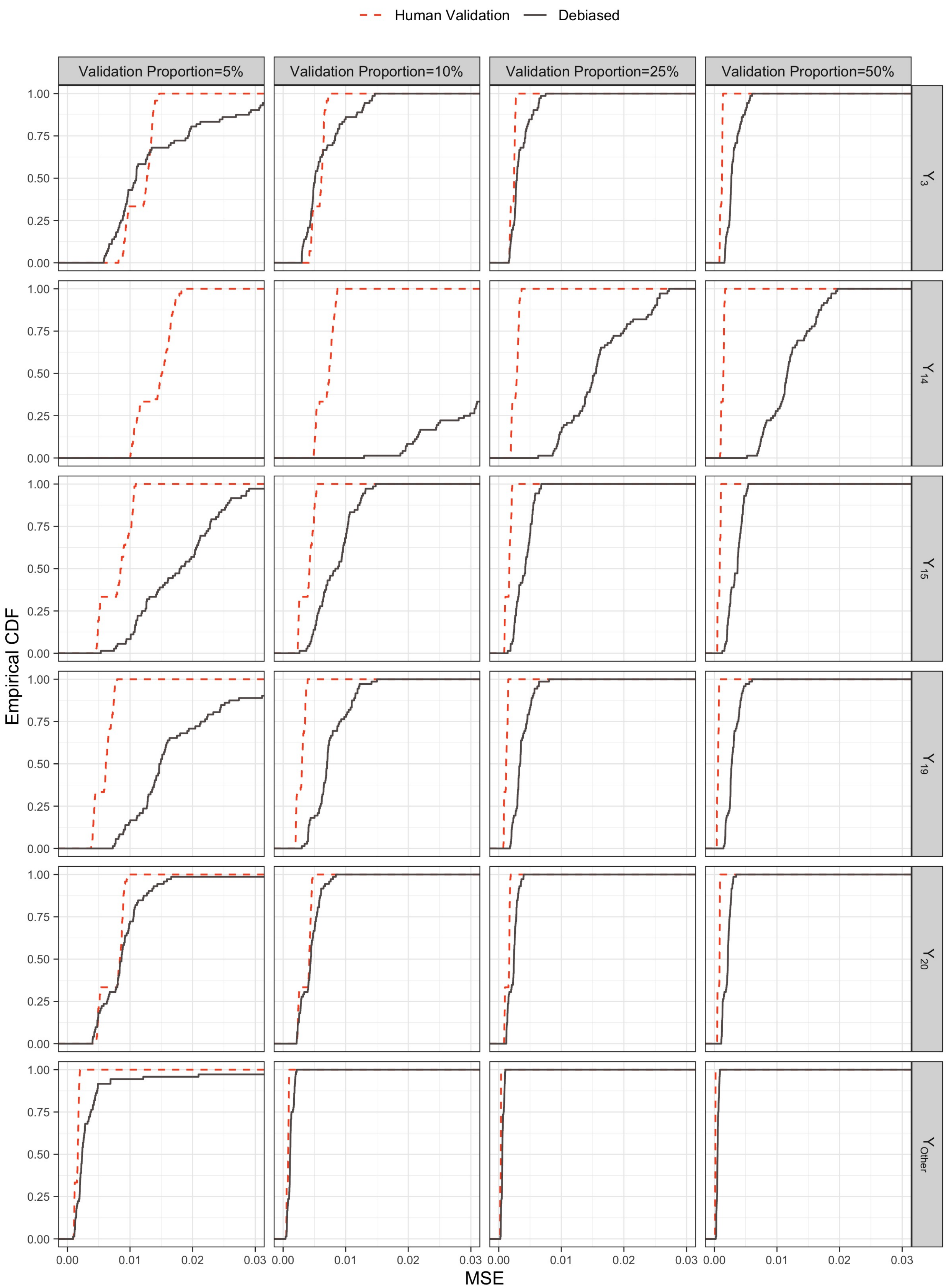


Figure 18: MSE Empirical CDF of $\beta_3$ by Model and Proportion of Validation Samples. Proxy on the RHS.

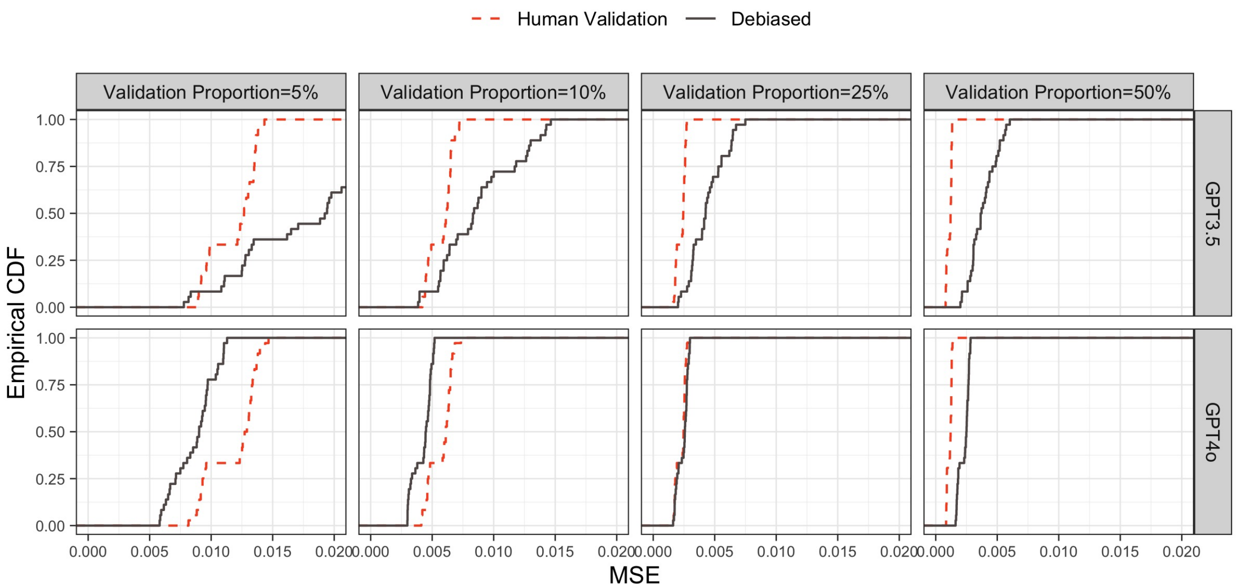



Figure 19: MSE Empirical CDF of $\beta_{14}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

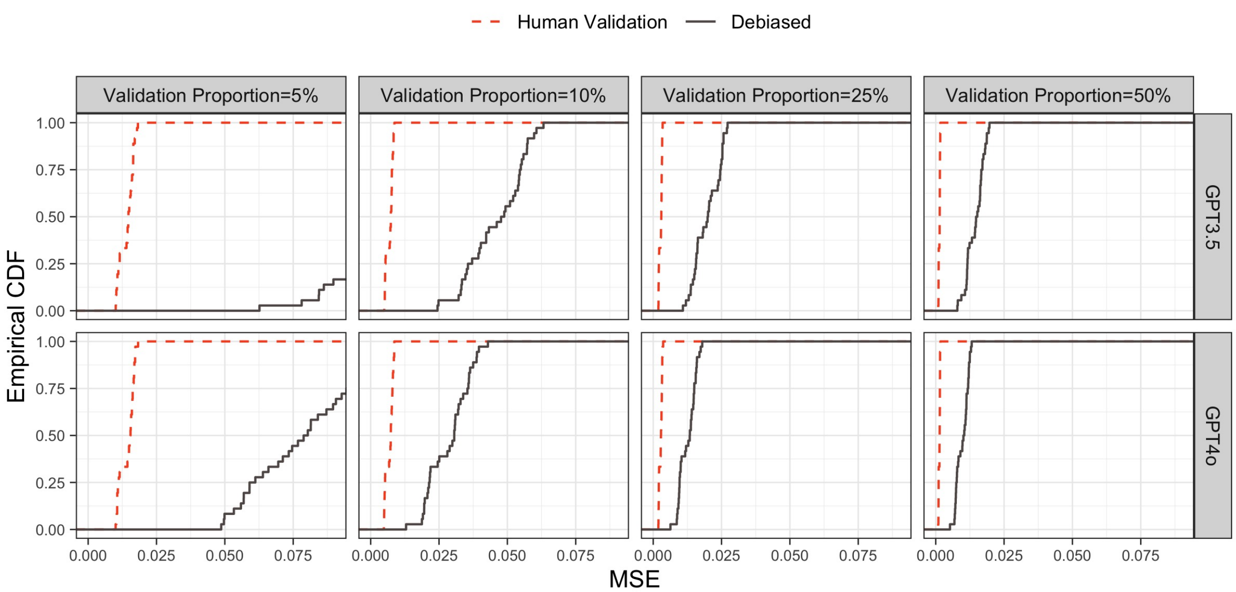


Figure 20: MSE Empirical CDF of $\beta_{15}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

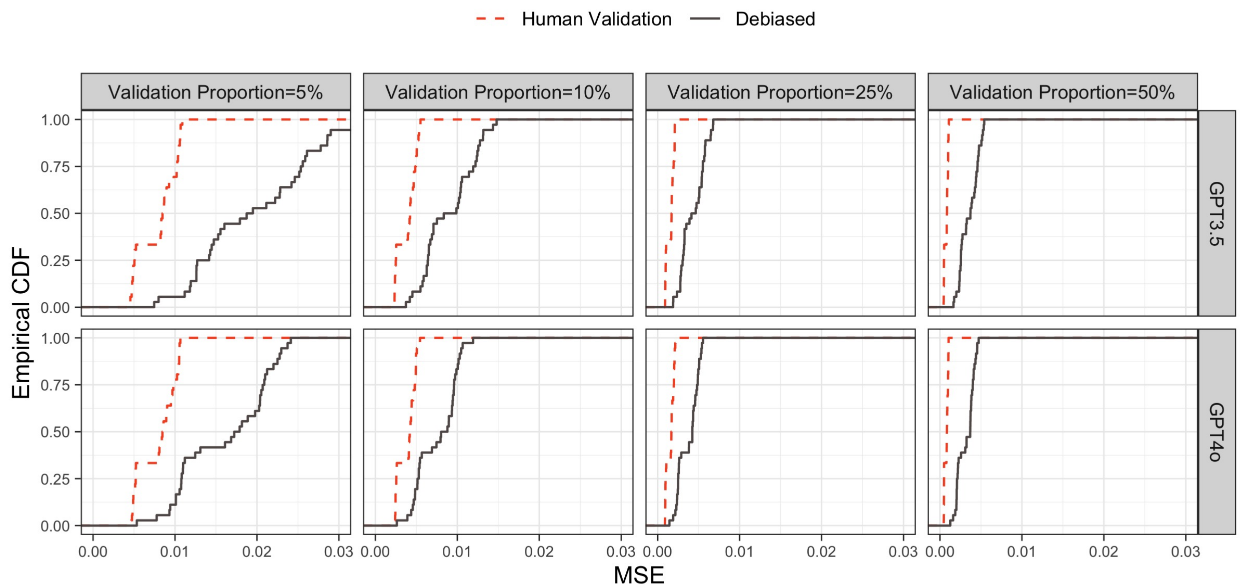



Figure 21: MSE Empirical CDF of $\beta_{19}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

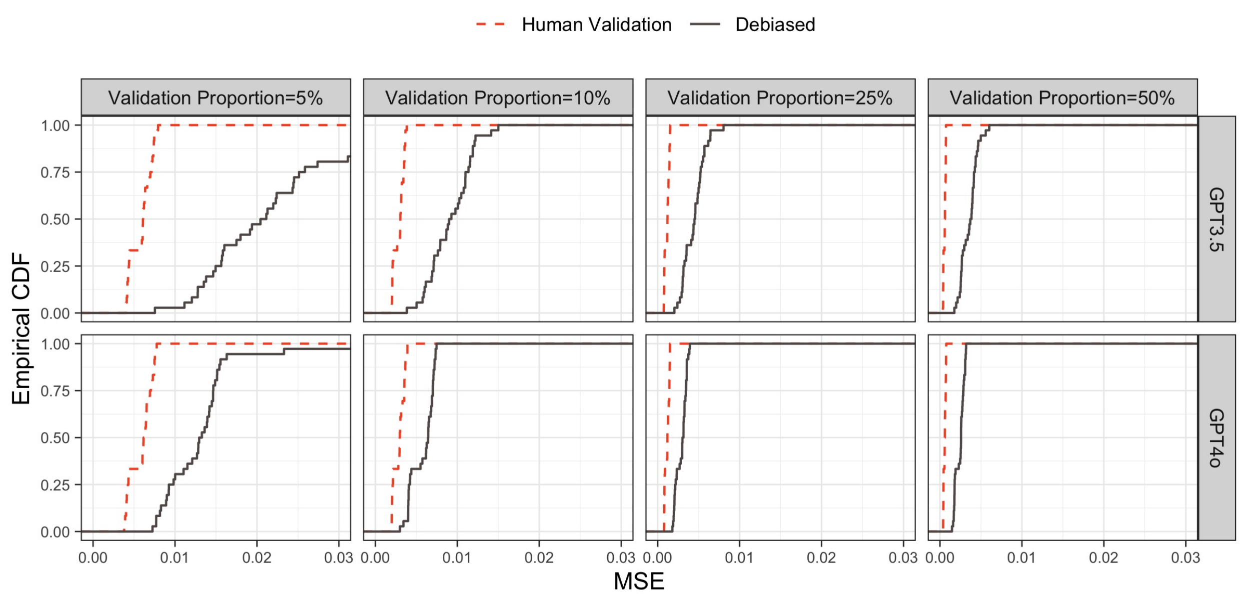


Figure 22: MSE Empirical CDF of $\beta_{20}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

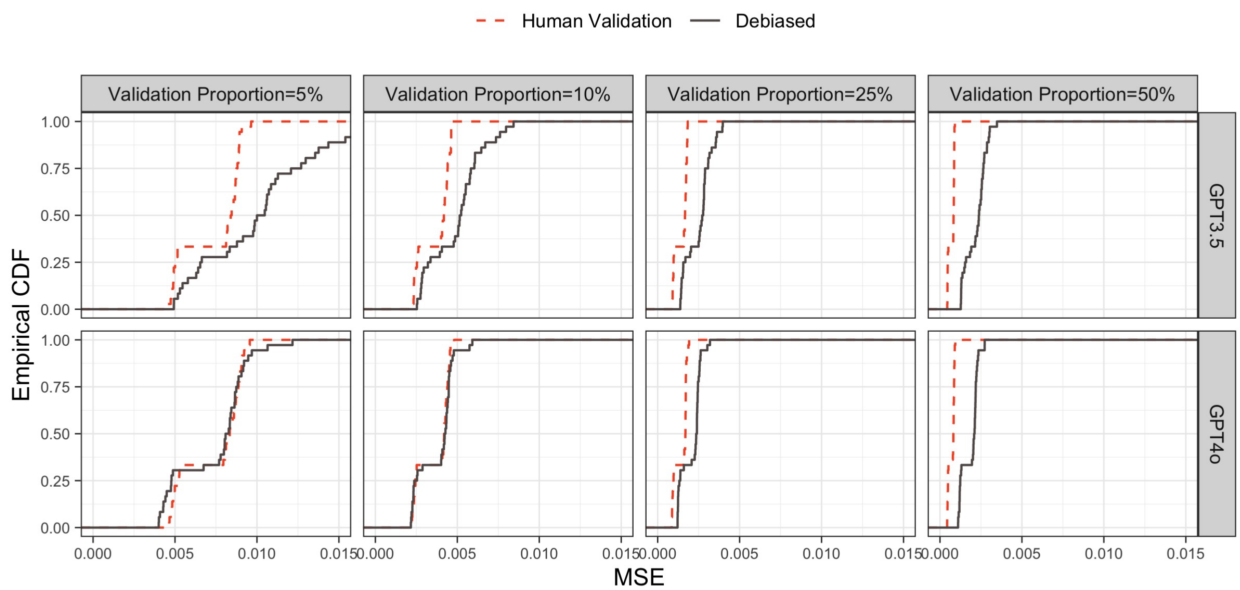



Figure 23: MSE Empirical CDF of $\beta_{\text{Other}}$ by Model and Proportion of Validation Samples. Proxy on the RHS.

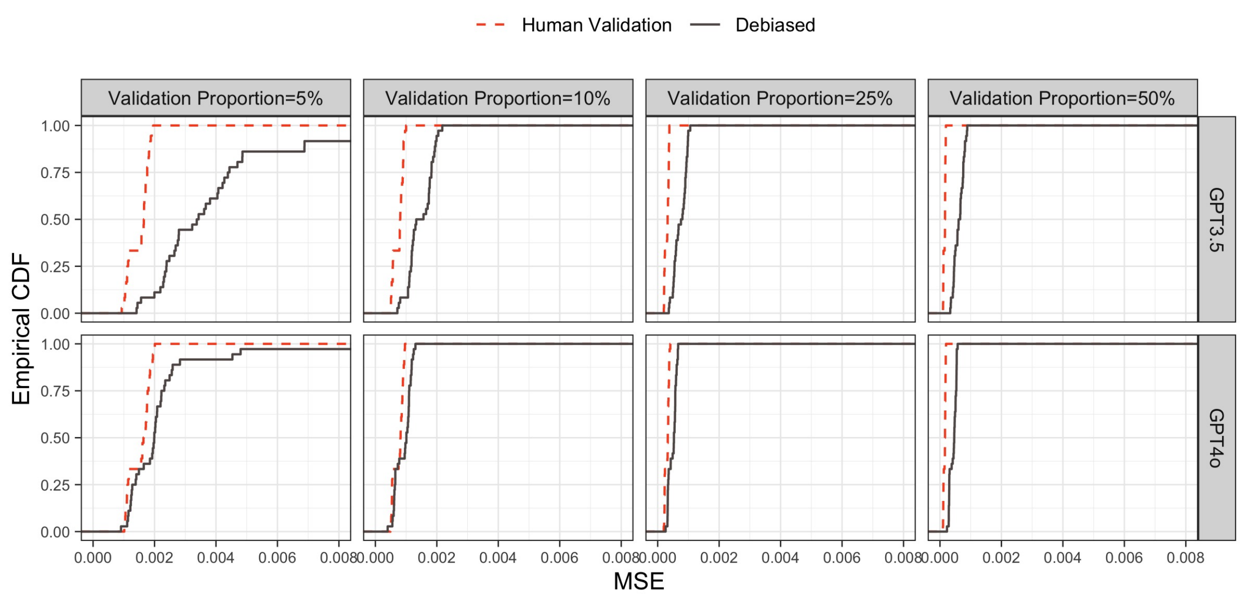


Table 1: Bias Summary Statistics of $\beta_3$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | 0.003 | 0.021 | 0.004 | -0.035 | 0.035 |
| 5% | Human Validation | 0.000 | 0.003 | 0.001 | -0.005 | 0.006 |
| 5% | Debiased | 0.001 | 0.004 | 0.000 | -0.005 | 0.006 |
| 10% | LLM | 0.003 | 0.022 | 0.003 | -0.035 | 0.035 |
| 10% | Human Validation | 0.000 | 0.002 | 0.000 | -0.004 | 0.004 |
| 10% | Debiased | 0.000 | 0.003 | 0.000 | -0.004 | 0.004 |
| 25% | LLM | 0.003 | 0.022 | 0.004 | -0.036 | 0.035 |
| 25% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| 25% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| 50% | LLM | 0.003 | 0.022 | 0.003 | -0.036 | 0.036 |
| 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| 50% | Debiased | 0.000 | 0.002 | 0.000 | -0.002 | 0.003 |

Table 2: Bias Summary Statistics of $\beta_{14}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | -0.003 | 0.031 | -0.012 | -0.043 | 0.049 |
| 5% | Human Validation | 0.000 | 0.003 | 0.000 | -0.006 | 0.005 |
| 5% | Debiased | 0.007 | 0.063 | 0.000 | -0.030 | 0.042 |
| 10% | LLM | -0.003 | 0.031 | -0.012 | -0.043 | 0.050 |
| 10% | Human Validation | 0.000 | 0.002 | -0.001 | -0.004 | 0.003 |
| 10% | Debiased | 0.001 | 0.008 | 0.001 | -0.010 | 0.013 |
| 25% | LLM | -0.003 | 0.031 | -0.012 | -0.043 | 0.050 |
| 25% | Human Validation | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| 25% | Debiased | 0.000 | 0.004 | 0.001 | -0.007 | 0.006 |
| 50% | LLM | -0.003 | 0.031 | -0.012 | -0.043 | 0.048 |
| 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.001 | 0.002 |
| 50% | Debiased | 0.000 | 0.004 | 0.000 | -0.005 | 0.006 |

Table 3: Bias Summary Statistics of $\beta_{15}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | 0.003 | 0.011 | 0.003 | -0.017 | 0.020 |
| 5% | Human Validation | 0.000 | 0.003 | 0.000 | -0.005 | 0.004 |
| 5% | Debiased | 0.000 | 0.004 | 0.000 | -0.007 | 0.006 |
| 10% | LLM | 0.002 | 0.011 | 0.003 | -0.018 | 0.020 |
| 10% | Human Validation | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| 10% | Debiased | 0.000 | 0.003 | 0.000 | -0.005 | 0.004 |
| 25% | LLM | 0.003 | 0.011 | 0.003 | -0.017 | 0.020 |
| 25% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| 25% | Debiased | 0.000 | 0.002 | 0.001 | -0.003 | 0.003 |
| 50% | LLM | 0.003 | 0.011 | 0.003 | -0.017 | 0.019 |
| 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.001 |
| 50% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |

Table 4: Bias Summary Statistics of $\beta_{19}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---:|---|---:|---:|---:|---:|---:|
| 5% | LLM | 0.007 | 0.012 | 0.009 | -0.011 | 0.023 |
| 5% | Human Validation | 0.001 | 0.003 | 0.001 | -0.004 | 0.005 |
| 5% | Debiased | -0.002 | 0.009 | -0.001 | -0.010 | 0.005 |
| 10% | LLM | 0.007 | 0.012 | 0.009 | -0.011 | 0.024 |
| 10% | Human Validation | 0.000 | 0.002 | 0.000 | -0.003 | 0.002 |
| 10% | Debiased | 0.000 | 0.003 | 0.000 | -0.005 | 0.004 |
| 25% | LLM | 0.006 | 0.012 | 0.009 | -0.011 | 0.024 |
| 25% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| 25% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| 50% | LLM | 0.006 | 0.012 | 0.009 | -0.011 | 0.024 |
| 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.001 | 0.001 |
| 50% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |

Table 5: Bias Summary Statistics of $\beta_{20}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---:|---|---:|---:|---:|---:|---:|
| 5% | LLM | -0.001 | 0.013 | 0.001 | -0.022 | 0.016 |
| 5% | Human Validation | 0.000 | 0.003 | 0.000 | -0.005 | 0.004 |
| 5% | Debiased | 0.000 | 0.003 | 0.000 | -0.004 | 0.005 |
| 10% | LLM | -0.001 | 0.013 | 0.001 | -0.023 | 0.015 |
| 10% | Human Validation | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| 10% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.004 |
| 25% | LLM | -0.001 | 0.013 | 0.001 | -0.021 | 0.015 |
| 25% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| 25% | Debiased | 0.000 | 0.001 | 0.000 | -0.002 | 0.003 |
| 50% | LLM | -0.001 | 0.012 | 0.001 | -0.021 | 0.015 |
| 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.001 | 0.002 |
| 50% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |

Table 6: Bias Summary Statistics of $\beta_{\text{Other}}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---:|---|---:|---:|---:|---:|---:|
| 5% | LLM | 0 | 0.008 | 0 | -0.012 | 0.013 |
| 5% | Human Validation | 0 | 0.001 | 0 | -0.002 | 0.002 |
| 5% | Debiased | 0 | 0.003 | 0 | -0.003 | 0.003 |
| 10% | LLM | 0 | 0.008 | 0 | -0.012 | 0.013 |
| 10% | Human Validation | 0 | 0.001 | 0 | -0.001 | 0.001 |
| 10% | Debiased | 0 | 0.001 | 0 | -0.002 | 0.002 |
| 25% | LLM | 0 | 0.008 | 0 | -0.012 | 0.014 |
| 25% | Human Validation | 0 | 0.001 | 0 | -0.001 | 0.001 |
| 25% | Debiased | 0 | 0.001 | 0 | -0.002 | 0.001 |
| 50% | LLM | 0 | 0.008 | 0 | -0.012 | 0.013 |
| 50% | Human Validation | 0 | 0.000 | 0 | -0.001 | 0.001 |
| 50% | Debiased | 0 | 0.001 | 0 | -0.001 | 0.001 |

Table 7: Normalized Bias Summary Statistics of $\beta_3$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | 0.148 | 0.846 | 0.166 | -1.236 | 1.398 |
| 5% | Human Validation | 0.001 | 0.032 | 0.005 | -0.052 | 0.057 |
| 5% | Debiased | 0.003 | 0.033 | 0.003 | -0.051 | 0.051 |
| 10% | LLM | 0.150 | 0.842 | 0.139 | -1.202 | 1.419 |
| 10% | Human Validation | -0.002 | 0.029 | -0.006 | -0.051 | 0.049 |
| 10% | Debiased | 0.000 | 0.033 | -0.003 | -0.056 | 0.050 |
| 25% | LLM | 0.154 | 0.852 | 0.165 | -1.287 | 1.419 |
| 25% | Human Validation | 0.006 | 0.028 | 0.008 | -0.040 | 0.047 |
| 25% | Debiased | -0.001 | 0.035 | -0.001 | -0.055 | 0.057 |
| 50% | LLM | 0.150 | 0.845 | 0.124 | -1.286 | 1.444 |
| 50% | Human Validation | 0.001 | 0.032 | 0.003 | -0.053 | 0.045 |
| 50% | Debiased | -0.002 | 0.029 | -0.004 | -0.046 | 0.047 |

Table 8: Normalized Bias Summary Statistics of $\beta_{14}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | 0.013 | 1.541 | -0.537 | -1.885 | 2.673 |
| 5% | Human Validation | -0.002 | 0.029 | -0.001 | -0.050 | 0.044 |
| 5% | Debiased | 0.004 | 0.039 | 0.002 | -0.057 | 0.063 |
| 10% | LLM | 0.013 | 1.543 | -0.553 | -1.929 | 2.667 |
| 10% | Human Validation | -0.006 | 0.028 | -0.006 | -0.046 | 0.043 |
| 10% | Debiased | 0.002 | 0.039 | 0.003 | -0.058 | 0.067 |
| 25% | LLM | 0.005 | 1.541 | -0.560 | -1.934 | 2.622 |
| 25% | Human Validation | 0.001 | 0.032 | -0.001 | -0.047 | 0.050 |
| 25% | Debiased | -0.001 | 0.034 | 0.005 | -0.059 | 0.056 |
| 50% | LLM | 0.009 | 1.547 | -0.543 | -1.968 | 2.667 |
| 50% | Human Validation | 0.003 | 0.032 | 0.004 | -0.042 | 0.050 |
| 50% | Debiased | 0.001 | 0.034 | 0.001 | -0.057 | 0.057 |

Table 9: Normalized Bias Summary Statistics of $\beta_{15}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | 0.118 | 0.463 | 0.166 | -0.633 | 0.791 |
| 5% | Human Validation | -0.003 | 0.033 | -0.004 | -0.056 | 0.050 |
| 5% | Debiased | -0.002 | 0.030 | -0.002 | -0.046 | 0.045 |
| 10% | LLM | 0.108 | 0.470 | 0.167 | -0.695 | 0.764 |
| 10% | Human Validation | -0.001 | 0.034 | 0.000 | -0.047 | 0.056 |
| 10% | Debiased | -0.003 | 0.033 | -0.002 | -0.060 | 0.048 |
| 25% | LLM | 0.121 | 0.460 | 0.150 | -0.653 | 0.782 |
| 25% | Human Validation | -0.005 | 0.034 | -0.012 | -0.055 | 0.056 |
| 25% | Debiased | 0.004 | 0.035 | 0.007 | -0.052 | 0.054 |
| 50% | LLM | 0.112 | 0.450 | 0.158 | -0.649 | 0.762 |
| 50% | Human Validation | -0.006 | 0.028 | -0.003 | -0.052 | 0.035 |
| 50% | Debiased | -0.002 | 0.030 | 0.000 | -0.051 | 0.042 |

Table 10: Normalized Bias Summary Statistics of $\beta_{19}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---:|---|---:|---:|---:|---:|---:|
| 5% | LLM | 0.369 | 0.615 | 0.486 | -0.534 | 1.285 |
| 5% | Human Validation | 0.008 | 0.034 | 0.010 | -0.050 | 0.062 |
| 5% | Debiased | -0.010 | 0.034 | -0.009 | -0.064 | 0.039 |
| 10% | LLM | 0.368 | 0.615 | 0.466 | -0.502 | 1.344 |
| 10% | Human Validation | -0.004 | 0.029 | -0.006 | -0.044 | 0.039 |
| 10% | Debiased | -0.004 | 0.034 | -0.003 | -0.064 | 0.048 |
| 25% | LLM | 0.359 | 0.607 | 0.451 | -0.493 | 1.326 |
| 25% | Human Validation | -0.001 | 0.032 | -0.001 | -0.058 | 0.055 |
| 25% | Debiased | 0.000 | 0.033 | -0.002 | -0.045 | 0.054 |
| 50% | LLM | 0.361 | 0.619 | 0.466 | -0.554 | 1.331 |
| 50% | Human Validation | 0.007 | 0.027 | 0.007 | -0.034 | 0.047 |
| 50% | Debiased | -0.002 | 0.035 | 0.000 | -0.069 | 0.049 |

Table 11: Normalized Bias Summary Statistics of $\beta_{20}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---:|---|---:|---:|---:|---:|---:|
| 5% | LLM | -0.008 | 0.632 | 0.023 | -1.000 | 0.888 |
| 5% | Human Validation | -0.001 | 0.034 | 0.001 | -0.059 | 0.047 |
| 5% | Debiased | 0.002 | 0.030 | 0.001 | -0.041 | 0.048 |
| 10% | LLM | -0.002 | 0.630 | 0.041 | -1.005 | 0.864 |
| 10% | Human Validation | 0.001 | 0.030 | 0.002 | -0.049 | 0.043 |
| 10% | Debiased | 0.004 | 0.033 | 0.004 | -0.048 | 0.056 |
| 25% | LLM | -0.009 | 0.623 | 0.029 | -0.970 | 0.838 |
| 25% | Human Validation | 0.003 | 0.030 | 0.004 | -0.049 | 0.047 |
| 25% | Debiased | -0.001 | 0.030 | -0.007 | -0.040 | 0.060 |
| 50% | LLM | -0.003 | 0.619 | 0.049 | -0.990 | 0.835 |
| 50% | Human Validation | 0.001 | 0.034 | 0.000 | -0.055 | 0.054 |
| 50% | Debiased | 0.000 | 0.036 | 0.003 | -0.055 | 0.061 |

Table 12: Normalized Bias Summary Statistics of $\beta_{\text{Other}}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---:|---|---:|---:|---:|---:|---:|
| 5% | LLM | -0.093 | 0.979 | 0.030 | -1.766 | 1.413 |
| 5% | Human Validation | -0.003 | 0.033 | -0.006 | -0.051 | 0.055 |
| 5% | Debiased | 0.001 | 0.031 | -0.006 | -0.046 | 0.055 |
| 10% | LLM | -0.100 | 0.961 | 0.021 | -1.788 | 1.394 |
| 10% | Human Validation | -0.001 | 0.030 | 0.001 | -0.049 | 0.044 |
| 10% | Debiased | 0.000 | 0.035 | -0.004 | -0.049 | 0.060 |
| 25% | LLM | -0.100 | 0.970 | 0.016 | -1.767 | 1.439 |
| 25% | Human Validation | 0.001 | 0.030 | -0.002 | -0.045 | 0.053 |
| 25% | Debiased | -0.005 | 0.036 | -0.004 | -0.065 | 0.055 |
| 50% | LLM | -0.082 | 0.957 | 0.036 | -1.721 | 1.424 |
| 50% | Human Validation | -0.001 | 0.034 | 0.008 | -0.057 | 0.043 |
| 50% | Debiased | 0.004 | 0.037 | 0.001 | -0.050 | 0.060 |

Table 13: Bias Summary Statistics of $\beta_3$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.003 | 0.021 | 0.004 | -0.035 | 0.035 |
| NA | 5% | Human Validation | 0.000 | 0.003 | 0.001 | -0.005 | 0.006 |
| NA | 5% | Debiased | 0.001 | 0.004 | 0.000 | -0.005 | 0.006 |
| NA | 10% | LLM | 0.003 | 0.022 | 0.003 | -0.035 | 0.035 |
| NA | 10% | Human Validation | 0.000 | 0.002 | 0.000 | -0.004 | 0.004 |
| NA | 10% | Debiased | 0.000 | 0.003 | 0.000 | -0.004 | 0.004 |
| NA | 25% | LLM | 0.003 | 0.022 | 0.004 | -0.036 | 0.035 |
| NA | 25% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| NA | 25% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| NA | 50% | LLM | 0.003 | 0.022 | 0.003 | -0.036 | 0.036 |
| NA | 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| NA | 50% | Debiased | 0.000 | 0.002 | 0.000 | -0.002 | 0.003 |

Table 14: Bias Summary Statistics of $\beta_{14}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | -0.003 | 0.031 | -0.012 | -0.043 | 0.049 |
| NA | 5% | Human Validation | 0.000 | 0.003 | 0.000 | -0.006 | 0.005 |
| NA | 5% | Debiased | 0.007 | 0.063 | 0.000 | -0.030 | 0.042 |
| NA | 10% | LLM | -0.003 | 0.031 | -0.012 | -0.043 | 0.050 |
| NA | 10% | Human Validation | 0.000 | 0.002 | -0.001 | -0.004 | 0.003 |
| NA | 10% | Debiased | 0.001 | 0.008 | 0.001 | -0.010 | 0.013 |
| NA | 25% | LLM | -0.003 | 0.031 | -0.012 | -0.043 | 0.050 |
| NA | 25% | Human Validation | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| NA | 25% | Debiased | 0.000 | 0.004 | 0.001 | -0.007 | 0.006 |
| NA | 50% | LLM | -0.003 | 0.031 | -0.012 | -0.043 | 0.048 |
| NA | 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.001 | 0.002 |
| NA | 50% | Debiased | 0.000 | 0.004 | 0.000 | -0.005 | 0.006 |

Table 15: Bias Summary Statistics of $\beta_{15}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.003 | 0.011 | 0.003 | -0.017 | 0.020 |
| NA | 5% | Human Validation | 0.000 | 0.003 | 0.000 | -0.005 | 0.004 |
| NA | 5% | Debiased | 0.000 | 0.004 | 0.000 | -0.007 | 0.006 |
| NA | 10% | LLM | 0.002 | 0.011 | 0.003 | -0.018 | 0.020 |
| NA | 10% | Human Validation | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| NA | 10% | Debiased | 0.000 | 0.003 | 0.000 | -0.005 | 0.004 |
| NA | 25% | LLM | 0.003 | 0.011 | 0.003 | -0.017 | 0.020 |
| NA | 25% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| NA | 25% | Debiased | 0.000 | 0.002 | 0.001 | -0.003 | 0.003 |
| NA | 50% | LLM | 0.003 | 0.011 | 0.003 | -0.017 | 0.019 |
| NA | 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.001 |
| NA | 50% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |

Table 16: Bias Summary Statistics of $\beta_{19}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.007 | 0.012 | 0.009 | -0.011 | 0.023 |
| NA | 5% | Human Validation | 0.001 | 0.003 | 0.001 | -0.004 | 0.005 |
| NA | 5% | Debiased | -0.002 | 0.009 | -0.001 | -0.010 | 0.005 |
| NA | 10% | LLM | 0.007 | 0.012 | 0.009 | -0.011 | 0.024 |
| NA | 10% | Human Validation | 0.000 | 0.002 | 0.000 | -0.003 | 0.002 |
| NA | 10% | Debiased | 0.000 | 0.003 | 0.000 | -0.005 | 0.004 |
| NA | 25% | LLM | 0.006 | 0.012 | 0.009 | -0.011 | 0.024 |
| NA | 25% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| NA | 25% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| NA | 50% | LLM | 0.006 | 0.012 | 0.009 | -0.011 | 0.024 |
| NA | 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.001 | 0.001 |
| NA | 50% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |

Table 17: Bias Summary Statistics of $\beta_{20}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | -0.001 | 0.013 | 0.001 | -0.022 | 0.016 |
| NA | 5% | Human Validation | 0.000 | 0.003 | 0.000 | -0.005 | 0.004 |
| NA | 5% | Debiased | 0.000 | 0.003 | 0.000 | -0.004 | 0.005 |
| NA | 10% | LLM | -0.001 | 0.013 | 0.001 | -0.023 | 0.015 |
| NA | 10% | Human Validation | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |
| NA | 10% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.004 |
| NA | 25% | LLM | -0.001 | 0.013 | 0.001 | -0.021 | 0.015 |
| NA | 25% | Human Validation | 0.000 | 0.001 | 0.000 | -0.002 | 0.002 |
| NA | 25% | Debiased | 0.000 | 0.001 | 0.000 | -0.002 | 0.003 |
| NA | 50% | LLM | -0.001 | 0.012 | 0.001 | -0.021 | 0.015 |
| NA | 50% | Human Validation | 0.000 | 0.001 | 0.000 | -0.001 | 0.002 |
| NA | 50% | Debiased | 0.000 | 0.002 | 0.000 | -0.003 | 0.003 |

Table 18: Bias Summary Statistics of $\beta_{\text{Other}}$ by Model and Proportion of Validation Samples for. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0 | 0.008 | 0 | -0.012 | 0.013 |
| NA | 5% | Human Validation | 0 | 0.001 | 0 | -0.002 | 0.002 |
| NA | 5% | Debiased | 0 | 0.003 | 0 | -0.003 | 0.003 |
| NA | 10% | LLM | 0 | 0.008 | 0 | -0.012 | 0.013 |
| NA | 10% | Human Validation | 0 | 0.001 | 0 | -0.001 | 0.001 |
| NA | 10% | Debiased | 0 | 0.001 | 0 | -0.002 | 0.002 |
| NA | 25% | LLM | 0 | 0.008 | 0 | -0.012 | 0.014 |
| NA | 25% | Human Validation | 0 | 0.001 | 0 | -0.001 | 0.001 |
| NA | 25% | Debiased | 0 | 0.001 | 0 | -0.002 | 0.001 |
| NA | 50% | LLM | 0 | 0.008 | 0 | -0.012 | 0.013 |
| NA | 50% | Human Validation | 0 | 0.000 | 0 | -0.001 | 0.001 |
| NA | 50% | Debiased | 0 | 0.001 | 0 | -0.001 | 0.001 |

Table 19: Normalized Bias Summary of $\beta_3$ Statistics by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.148 | 0.846 | 0.166 | -1.236 | 1.398 |
| NA | 5% | Human Validation | 0.001 | 0.032 | 0.005 | -0.052 | 0.057 |
| NA | 5% | Debiased | 0.003 | 0.033 | 0.003 | -0.051 | 0.051 |
| NA | 10% | LLM | 0.150 | 0.842 | 0.139 | -1.202 | 1.419 |
| NA | 10% | Human Validation | -0.002 | 0.029 | -0.006 | -0.051 | 0.049 |
| NA | 10% | Debiased | 0.000 | 0.033 | -0.003 | -0.056 | 0.050 |
| NA | 25% | LLM | 0.154 | 0.852 | 0.165 | -1.287 | 1.419 |
| NA | 25% | Human Validation | 0.006 | 0.028 | 0.008 | -0.040 | 0.047 |
| NA | 25% | Debiased | -0.001 | 0.035 | -0.001 | -0.055 | 0.057 |
| NA | 50% | LLM | 0.150 | 0.845 | 0.124 | -1.286 | 1.444 |
| NA | 50% | Human Validation | 0.001 | 0.032 | 0.003 | -0.053 | 0.045 |
| NA | 50% | Debiased | -0.002 | 0.029 | -0.004 | -0.046 | 0.047 |

Table 20: Normalized Bias Summary Statistics of $\beta_{14}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.013 | 1.541 | -0.537 | -1.885 | 2.673 |
| NA | 5% | Human Validation | -0.002 | 0.029 | -0.001 | -0.050 | 0.044 |
| NA | 5% | Debiased | 0.004 | 0.039 | 0.002 | -0.057 | 0.063 |
| NA | 10% | LLM | 0.013 | 1.543 | -0.553 | -1.929 | 2.667 |
| NA | 10% | Human Validation | -0.006 | 0.028 | -0.006 | -0.046 | 0.043 |
| NA | 10% | Debiased | 0.002 | 0.039 | 0.003 | -0.058 | 0.067 |
| NA | 25% | LLM | 0.005 | 1.541 | -0.560 | -1.934 | 2.622 |
| NA | 25% | Human Validation | 0.001 | 0.032 | -0.001 | -0.047 | 0.050 |
| NA | 25% | Debiased | -0.001 | 0.034 | 0.005 | -0.059 | 0.056 |
| NA | 50% | LLM | 0.009 | 1.547 | -0.543 | -1.968 | 2.667 |
| NA | 50% | Human Validation | 0.003 | 0.032 | 0.004 | -0.042 | 0.050 |
| NA | 50% | Debiased | 0.001 | 0.034 | 0.001 | -0.057 | 0.057 |

Table 21: Normalized Bias Summary Statistics of $\beta_{15}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.118 | 0.463 | 0.166 | -0.633 | 0.791 |
| NA | 5% | Human Validation | -0.003 | 0.033 | -0.004 | -0.056 | 0.050 |
| NA | 5% | Debiased | -0.002 | 0.030 | -0.002 | -0.046 | 0.045 |
| NA | 10% | LLM | 0.108 | 0.470 | 0.167 | -0.695 | 0.764 |
| NA | 10% | Human Validation | -0.001 | 0.034 | 0.000 | -0.047 | 0.056 |
| NA | 10% | Debiased | -0.003 | 0.033 | -0.002 | -0.060 | 0.048 |
| NA | 25% | LLM | 0.121 | 0.460 | 0.150 | -0.653 | 0.782 |
| NA | 25% | Human Validation | -0.005 | 0.034 | -0.012 | -0.055 | 0.056 |
| NA | 25% | Debiased | 0.004 | 0.035 | 0.007 | -0.052 | 0.054 |
| NA | 50% | LLM | 0.112 | 0.450 | 0.158 | -0.649 | 0.762 |
| NA | 50% | Human Validation | -0.006 | 0.028 | -0.003 | -0.052 | 0.035 |
| NA | 50% | Debiased | -0.002 | 0.030 | 0.000 | -0.051 | 0.042 |

Table 22: Normalized Bias Summary Statistics of $\beta_{19}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.369 | 0.615 | 0.486 | -0.534 | 1.285 |
| NA | 5% | Human Validation | 0.008 | 0.034 | 0.010 | -0.050 | 0.062 |
| NA | 5% | Debiased | -0.010 | 0.034 | -0.009 | -0.064 | 0.039 |
| NA | 10% | LLM | 0.368 | 0.615 | 0.466 | -0.502 | 1.344 |
| NA | 10% | Human Validation | -0.004 | 0.029 | -0.006 | -0.044 | 0.039 |
| NA | 10% | Debiased | -0.004 | 0.034 | -0.003 | -0.064 | 0.048 |
| NA | 25% | LLM | 0.359 | 0.607 | 0.451 | -0.493 | 1.326 |
| NA | 25% | Human Validation | -0.001 | 0.032 | -0.001 | -0.058 | 0.055 |
| NA | 25% | Debiased | 0.000 | 0.033 | -0.002 | -0.045 | 0.054 |
| NA | 50% | LLM | 0.361 | 0.619 | 0.466 | -0.554 | 1.331 |
| NA | 50% | Human Validation | 0.007 | 0.027 | 0.007 | -0.034 | 0.047 |
| NA | 50% | Debiased | -0.002 | 0.035 | 0.000 | -0.069 | 0.049 |

Table 23: Normalized Bias Summary Statistics of $\beta_{20}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | -0.008 | 0.632 | 0.023 | -1.000 | 0.888 |
| NA | 5% | Human Validation | -0.001 | 0.034 | 0.001 | -0.059 | 0.047 |
| NA | 5% | Debiased | 0.002 | 0.030 | 0.001 | -0.041 | 0.048 |
| NA | 10% | LLM | -0.002 | 0.630 | 0.041 | -1.005 | 0.864 |
| NA | 10% | Human Validation | 0.001 | 0.030 | 0.002 | -0.049 | 0.043 |
| NA | 10% | Debiased | 0.004 | 0.033 | 0.004 | -0.048 | 0.056 |
| NA | 25% | LLM | -0.009 | 0.623 | 0.029 | -0.970 | 0.838 |
| NA | 25% | Human Validation | 0.003 | 0.030 | 0.004 | -0.049 | 0.047 |
| NA | 25% | Debiased | -0.001 | 0.030 | -0.007 | -0.040 | 0.060 |
| NA | 50% | LLM | -0.003 | 0.619 | 0.049 | -0.990 | 0.835 |
| NA | 50% | Human Validation | 0.001 | 0.034 | 0.000 | -0.055 | 0.054 |
| NA | 50% | Debiased | 0.000 | 0.036 | 0.003 | -0.055 | 0.061 |

Table 24: Normalized Bias Summary Statistics of $\beta_{\text{Other}}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | Normalized Bias | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | -0.093 | 0.979 | 0.030 | -1.766 | 1.413 |
| NA | 5% | Human Validation | -0.003 | 0.033 | -0.006 | -0.051 | 0.055 |
| NA | 5% | Debiased | 0.001 | 0.031 | -0.006 | -0.046 | 0.055 |
| NA | 10% | LLM | -0.100 | 0.961 | 0.021 | -1.788 | 1.394 |
| NA | 10% | Human Validation | -0.001 | 0.030 | 0.001 | -0.049 | 0.044 |
| NA | 10% | Debiased | 0.000 | 0.035 | -0.004 | -0.049 | 0.060 |
| NA | 25% | LLM | -0.100 | 0.970 | 0.016 | -1.767 | 1.439 |
| NA | 25% | Human Validation | 0.001 | 0.030 | -0.002 | -0.045 | 0.053 |
| NA | 25% | Debiased | -0.005 | 0.036 | -0.004 | -0.065 | 0.055 |
| NA | 50% | LLM | -0.082 | 0.957 | 0.036 | -1.721 | 1.424 |
| NA | 50% | Human Validation | -0.001 | 0.034 | 0.008 | -0.057 | 0.043 |
| NA | 50% | Debiased | 0.004 | 0.037 | 0.001 | -0.050 | 0.060 |

Table 25: MSE Summary Statistics of $\beta_3$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.002 |
| 5% | Human Validation | 0.012 | 0.002 | 0.013 | 0.009 | 0.014 |
| 5% | Debiased | 0.015 | 0.015 | 0.011 | 0.006 | 0.031 |
| 10% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.002 |
| 10% | Human Validation | 0.006 | 0.001 | 0.006 | 0.004 | 0.007 |
| 10% | Debiased | 0.007 | 0.003 | 0.005 | 0.003 | 0.013 |
| 25% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| 25% | Human Validation | 0.002 | 0.000 | 0.002 | 0.002 | 0.003 |
| 25% | Debiased | 0.003 | 0.001 | 0.003 | 0.002 | 0.006 |
| 50% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.002 |
| 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.001 | 0.001 |
| 50% | Debiased | 0.003 | 0.001 | 0.003 | 0.002 | 0.005 |

Table 26: MSE Summary Statistics of $\beta_{14}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| 5% | Human Validation | 0.014 | 0.003 | 0.015 | 0.010 | 0.017 |
| 5% | Debiased | 3.543 | 23.671 | 0.099 | 0.055 | 2.287 |
| 10% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| 10% | Human Validation | 0.007 | 0.001 | 0.007 | 0.005 | 0.009 |
| 10% | Debiased | 0.037 | 0.013 | 0.036 | 0.020 | 0.057 |
| 25% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| 25% | Human Validation | 0.003 | 0.001 | 0.003 | 0.002 | 0.003 |
| 25% | Debiased | 0.016 | 0.005 | 0.015 | 0.009 | 0.026 |
| 50% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.001 | 0.002 |
| 50% | Debiased | 0.012 | 0.004 | 0.012 | 0.007 | 0.018 |

Table 27: MSE Summary Statistics of $\beta_{15}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 5% | Human Validation | 0.008 | 0.002 | 0.009 | 0.005 | 0.011 |
| 5% | Debiased | 0.018 | 0.006 | 0.018 | 0.009 | 0.029 |
| 10% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 10% | Human Validation | 0.004 | 0.001 | 0.004 | 0.002 | 0.005 |
| 10% | Debiased | 0.008 | 0.003 | 0.009 | 0.004 | 0.013 |
| 25% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 25% | Human Validation | 0.002 | 0.000 | 0.002 | 0.001 | 0.002 |
| 25% | Debiased | 0.004 | 0.001 | 0.004 | 0.002 | 0.006 |
| 50% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 50% | Debiased | 0.003 | 0.001 | 0.004 | 0.002 | 0.005 |

Table 28: MSE Summary Statistics of $\beta_{19}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 5% | Human Validation | 0.006 | 0.001 | 0.006 | 0.004 | 0.008 |
| 5% | Debiased | 0.061 | 0.369 | 0.015 | 0.008 | 0.038 |
| 10% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 10% | Human Validation | 0.003 | 0.001 | 0.003 | 0.002 | 0.004 |
| 10% | Debiased | 0.008 | 0.003 | 0.007 | 0.004 | 0.012 |
| 25% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 25% | Human Validation | 0.001 | 0.000 | 0.001 | 0.001 | 0.001 |
| 25% | Debiased | 0.004 | 0.001 | 0.003 | 0.002 | 0.006 |
| 50% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 50% | Debiased | 0.003 | 0.001 | 0.003 | 0.002 | 0.005 |

Table 29: MSE Summary Statistics of $\beta_{20}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 5% | Human Validation | 0.007 | 0.002 | 0.008 | 0.005 | 0.009 |
| 5% | Debiased | 0.009 | 0.006 | 0.009 | 0.004 | 0.015 |
| 10% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 10% | Human Validation | 0.004 | 0.001 | 0.004 | 0.002 | 0.005 |
| 10% | Debiased | 0.004 | 0.001 | 0.004 | 0.002 | 0.007 |
| 25% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 25% | Human Validation | 0.001 | 0.000 | 0.002 | 0.001 | 0.002 |
| 25% | Debiased | 0.002 | 0.001 | 0.002 | 0.001 | 0.004 |
| 50% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 50% | Debiased | 0.002 | 0.001 | 0.002 | 0.001 | 0.003 |

Table 30: MSE Summary Statistics of $\beta_{\text{Other}}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|
| 5% | LLM | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 5% | Human Validation | 0.002 | 0.000 | 0.002 | 0.001 | 0.002 |
| 5% | Debiased | 0.011 | 0.052 | 0.002 | 0.001 | 0.009 |
| 10% | LLM | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 10% | Human Validation | 0.001 | 0.000 | 0.001 | 0.001 | 0.001 |
| 10% | Debiased | 0.001 | 0.000 | 0.001 | 0.001 | 0.002 |
| 25% | LLM | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 25% | Human Validation | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 25% | Debiased | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| 50% | LLM | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 50% | Human Validation | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| 50% | Debiased | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |

Table 31: MSE Summary Statistics of $\beta_3$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.002 |
| NA | 5% | Human Validation | 0.012 | 0.002 | 0.013 | 0.009 | 0.014 |
| NA | 5% | Debiased | 0.015 | 0.015 | 0.011 | 0.006 | 0.031 |
| NA | 10% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.002 |
| NA | 10% | Human Validation | 0.006 | 0.001 | 0.006 | 0.004 | 0.007 |
| NA | 10% | Debiased | 0.007 | 0.003 | 0.005 | 0.003 | 0.013 |
| NA | 25% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| NA | 25% | Human Validation | 0.002 | 0.000 | 0.002 | 0.002 | 0.003 |
| NA | 25% | Debiased | 0.003 | 0.001 | 0.003 | 0.002 | 0.006 |
| NA | 50% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.002 |
| NA | 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.001 | 0.001 |
| NA | 50% | Debiased | 0.003 | 0.001 | 0.003 | 0.002 | 0.005 |

Table 32: MSE Summary Statistics of $\beta_{14}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| NA | 5% | Human Validation | 0.014 | 0.003 | 0.015 | 0.010 | 0.017 |
| NA | 5% | Debiased | 3.543 | 23.671 | 0.099 | 0.055 | 2.287 |
| NA | 10% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| NA | 10% | Human Validation | 0.007 | 0.001 | 0.007 | 0.005 | 0.009 |
| NA | 10% | Debiased | 0.037 | 0.013 | 0.036 | 0.020 | 0.057 |
| NA | 25% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| NA | 25% | Human Validation | 0.003 | 0.001 | 0.003 | 0.002 | 0.003 |
| NA | 25% | Debiased | 0.016 | 0.005 | 0.015 | 0.009 | 0.026 |
| NA | 50% | LLM | 0.001 | 0.001 | 0.001 | 0.001 | 0.003 |
| NA | 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.001 | 0.002 |
| NA | 50% | Debiased | 0.012 | 0.004 | 0.012 | 0.007 | 0.018 |

Table 33: MSE Summary Statistics of $\beta_{15}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 5% | Human Validation | 0.008 | 0.002 | 0.009 | 0.005 | 0.011 |
| NA | 5% | Debiased | 0.018 | 0.006 | 0.018 | 0.009 | 0.029 |
| NA | 10% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 10% | Human Validation | 0.004 | 0.001 | 0.004 | 0.002 | 0.005 |
| NA | 10% | Debiased | 0.008 | 0.003 | 0.009 | 0.004 | 0.013 |
| NA | 25% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 25% | Human Validation | 0.002 | 0.000 | 0.002 | 0.001 | 0.002 |
| NA | 25% | Debiased | 0.004 | 0.001 | 0.004 | 0.002 | 0.006 |
| NA | 50% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 50% | Debiased | 0.003 | 0.001 | 0.004 | 0.002 | 0.005 |

Table 34: MSE Summary Statistics of $\beta_{19}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 5% | Human Validation | 0.006 | 0.001 | 0.006 | 0.004 | 0.008 |
| NA | 5% | Debiased | 0.061 | 0.369 | 0.015 | 0.008 | 0.038 |
| NA | 10% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 10% | Human Validation | 0.003 | 0.001 | 0.003 | 0.002 | 0.004 |
| NA | 10% | Debiased | 0.008 | 0.003 | 0.007 | 0.004 | 0.012 |
| NA | 25% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 25% | Human Validation | 0.001 | 0.000 | 0.001 | 0.001 | 0.001 |
| NA | 25% | Debiased | 0.004 | 0.001 | 0.003 | 0.002 | 0.006 |
| NA | 50% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 50% | Debiased | 0.003 | 0.001 | 0.003 | 0.002 | 0.005 |

Table 35: MSE Summary Statistics of $\beta_{20}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 5% | Human Validation | 0.007 | 0.002 | 0.008 | 0.005 | 0.009 |
| NA | 5% | Debiased | 0.009 | 0.006 | 0.009 | 0.004 | 0.015 |
| NA | 10% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 10% | Human Validation | 0.004 | 0.001 | 0.004 | 0.002 | 0.005 |
| NA | 10% | Debiased | 0.004 | 0.001 | 0.004 | 0.002 | 0.007 |
| NA | 25% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 25% | Human Validation | 0.001 | 0.000 | 0.002 | 0.001 | 0.002 |
| NA | 25% | Debiased | 0.002 | 0.001 | 0.002 | 0.001 | 0.004 |
| NA | 50% | LLM | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 50% | Human Validation | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 50% | Debiased | 0.002 | 0.001 | 0.002 | 0.001 | 0.003 |

Table 36: MSE Summary Statistics of $\beta_{\text{Other}}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| NA | 5% | Human Validation | 0.002 | 0.000 | 0.002 | 0.001 | 0.002 |
| NA | 5% | Debiased | 0.011 | 0.052 | 0.002 | 0.001 | 0.009 |
| NA | 10% | LLM | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| NA | 10% | Human Validation | 0.001 | 0.000 | 0.001 | 0.001 | 0.001 |
| NA | 10% | Debiased | 0.001 | 0.000 | 0.001 | 0.001 | 0.002 |
| NA | 25% | LLM | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| NA | 25% | Human Validation | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| NA | 25% | Debiased | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |
| NA | 50% | LLM | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| NA | 50% | Human Validation | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| NA | 50% | Debiased | 0.001 | 0.000 | 0.001 | 0.000 | 0.001 |

Table 37: Coverage Summary Statistics of $\beta_3$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.867 | 0.096 | 0.911 | 0.689 | 0.952 |
| 5% | Human Validation | 0.919 | 0.008 | 0.919 | 0.906 | 0.931 |
| 5% | Debiased | 0.920 | 0.011 | 0.921 | 0.904 | 0.939 |
| 10% | LLM | 0.866 | 0.097 | 0.913 | 0.679 | 0.947 |
| 10% | Human Validation | 0.935 | 0.008 | 0.935 | 0.923 | 0.947 |
| 10% | Debiased | 0.940 | 0.007 | 0.940 | 0.928 | 0.952 |
| 25% | LLM | 0.865 | 0.097 | 0.909 | 0.692 | 0.948 |
| 25% | Human Validation | 0.944 | 0.008 | 0.944 | 0.931 | 0.954 |
| 25% | Debiased | 0.945 | 0.007 | 0.946 | 0.934 | 0.955 |
| 50% | LLM | 0.865 | 0.099 | 0.913 | 0.653 | 0.947 |
| 50% | Human Validation | 0.949 | 0.007 | 0.948 | 0.939 | 0.959 |
| 50% | Debiased | 0.948 | 0.007 | 0.949 | 0.937 | 0.957 |

Table 38: Coverage Summary Statistics of $\beta_{14}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.685 | 0.225 | 0.764 | 0.252 | 0.932 |
| 5% | Human Validation | 0.917 | 0.009 | 0.917 | 0.901 | 0.930 |
| 5% | Debiased | 0.938 | 0.010 | 0.939 | 0.920 | 0.953 |
| 10% | LLM | 0.689 | 0.226 | 0.770 | 0.242 | 0.937 |
| 10% | Human Validation | 0.933 | 0.008 | 0.934 | 0.918 | 0.945 |
| 10% | Debiased | 0.940 | 0.008 | 0.942 | 0.926 | 0.951 |
| 25% | LLM | 0.686 | 0.225 | 0.759 | 0.237 | 0.938 |
| 25% | Human Validation | 0.943 | 0.008 | 0.944 | 0.929 | 0.955 |
| 25% | Debiased | 0.946 | 0.007 | 0.946 | 0.937 | 0.957 |
| 50% | LLM | 0.690 | 0.225 | 0.766 | 0.255 | 0.935 |
| 50% | Human Validation | 0.947 | 0.007 | 0.948 | 0.936 | 0.959 |
| 50% | Debiased | 0.948 | 0.007 | 0.949 | 0.937 | 0.958 |

Table 39: Coverage Summary Statistics of $\beta_{15}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.925 | 0.032 | 0.936 | 0.879 | 0.957 |
| 5% | Human Validation | 0.932 | 0.009 | 0.931 | 0.919 | 0.946 |
| 5% | Debiased | 0.932 | 0.010 | 0.933 | 0.919 | 0.950 |
| 10% | LLM | 0.926 | 0.030 | 0.938 | 0.876 | 0.953 |
| 10% | Human Validation | 0.940 | 0.008 | 0.941 | 0.927 | 0.951 |
| 10% | Debiased | 0.941 | 0.008 | 0.941 | 0.928 | 0.955 |
| 25% | LLM | 0.925 | 0.033 | 0.938 | 0.868 | 0.953 |
| 25% | Human Validation | 0.946 | 0.007 | 0.946 | 0.934 | 0.958 |
| 25% | Debiased | 0.947 | 0.007 | 0.947 | 0.934 | 0.957 |
| 50% | LLM | 0.926 | 0.031 | 0.933 | 0.883 | 0.956 |
| 50% | Human Validation | 0.947 | 0.006 | 0.947 | 0.938 | 0.958 |
| 50% | Debiased | 0.947 | 0.007 | 0.947 | 0.934 | 0.957 |

Table 40: Coverage Summary Statistics of $\beta_{19}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.891 | 0.065 | 0.915 | 0.740 | 0.954 |
| 5% | Human Validation | 0.937 | 0.009 | 0.938 | 0.926 | 0.953 |
| 5% | Debiased | 0.944 | 0.009 | 0.944 | 0.931 | 0.956 |
| 10% | LLM | 0.890 | 0.066 | 0.912 | 0.750 | 0.949 |
| 10% | Human Validation | 0.943 | 0.008 | 0.943 | 0.930 | 0.954 |
| 10% | Debiased | 0.947 | 0.008 | 0.947 | 0.934 | 0.957 |
| 25% | LLM | 0.894 | 0.064 | 0.915 | 0.737 | 0.954 |
| 25% | Human Validation | 0.947 | 0.007 | 0.947 | 0.938 | 0.958 |
| 25% | Debiased | 0.947 | 0.008 | 0.948 | 0.935 | 0.959 |
| 50% | LLM | 0.889 | 0.066 | 0.913 | 0.735 | 0.947 |
| 50% | Human Validation | 0.951 | 0.007 | 0.950 | 0.940 | 0.961 |
| 50% | Debiased | 0.947 | 0.007 | 0.948 | 0.935 | 0.957 |

Table 41: Coverage Summary Statistics of $\beta_{20}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.901 | 0.062 | 0.923 | 0.784 | 0.956 |
| 5% | Human Validation | 0.938 | 0.007 | 0.938 | 0.926 | 0.950 |
| 5% | Debiased | 0.938 | 0.008 | 0.938 | 0.926 | 0.951 |
| 10% | LLM | 0.902 | 0.056 | 0.920 | 0.785 | 0.953 |
| 10% | Human Validation | 0.943 | 0.007 | 0.942 | 0.932 | 0.956 |
| 10% | Debiased | 0.945 | 0.007 | 0.944 | 0.934 | 0.956 |
| 25% | LLM | 0.903 | 0.058 | 0.925 | 0.783 | 0.952 |
| 25% | Human Validation | 0.948 | 0.007 | 0.948 | 0.937 | 0.959 |
| 25% | Debiased | 0.947 | 0.007 | 0.947 | 0.937 | 0.959 |
| 50% | LLM | 0.904 | 0.055 | 0.923 | 0.796 | 0.954 |
| 50% | Human Validation | 0.950 | 0.007 | 0.951 | 0.938 | 0.960 |
| 50% | Debiased | 0.949 | 0.007 | 0.948 | 0.939 | 0.961 |

Table 42: Coverage Summary Statistics of $\beta_{\text{Other}}$ by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage | Mean | SD | Median | 5% | 95% |
|---:|---|---|---:|---:|---:|---:|
| 5% | LLM | 0.836 | 0.133 | 0.886 | 0.570 | 0.950 |
| 5% | Human Validation | 0.949 | 0.007 | 0.949 | 0.937 | 0.961 |
| 5% | Debiased | 0.954 | 0.007 | 0.954 | 0.943 | 0.966 |
| 10% | LLM | 0.841 | 0.133 | 0.891 | 0.550 | 0.955 |
| 10% | Human Validation | 0.950 | 0.007 | 0.950 | 0.939 | 0.960 |
| 10% | Debiased | 0.951 | 0.008 | 0.952 | 0.938 | 0.962 |
| 25% | LLM | 0.839 | 0.133 | 0.887 | 0.581 | 0.949 |
| 25% | Human Validation | 0.949 | 0.007 | 0.950 | 0.938 | 0.958 |
| 25% | Debiased | 0.948 | 0.007 | 0.950 | 0.935 | 0.959 |
| 50% | LLM | 0.840 | 0.129 | 0.891 | 0.561 | 0.951 |
| 50% | Human Validation | 0.951 | 0.007 | 0.950 | 0.940 | 0.962 |
| 50% | Debiased | 0.949 | 0.007 | 0.949 | 0.939 | 0.960 |

Table 43: Coverage Summary Statistics of $\beta_3$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.867 | 0.096 | 0.911 | 0.689 | 0.952 |
| NA | 5% | Human Validation | 0.919 | 0.008 | 0.919 | 0.906 | 0.931 |
| NA | 5% | Debiased | 0.920 | 0.011 | 0.921 | 0.904 | 0.939 |
| NA | 10% | LLM | 0.866 | 0.097 | 0.913 | 0.679 | 0.947 |
| NA | 10% | Human Validation | 0.935 | 0.008 | 0.935 | 0.923 | 0.947 |
| NA | 10% | Debiased | 0.940 | 0.007 | 0.940 | 0.928 | 0.952 |
| NA | 25% | LLM | 0.865 | 0.097 | 0.909 | 0.692 | 0.948 |
| NA | 25% | Human Validation | 0.944 | 0.008 | 0.944 | 0.931 | 0.954 |
| NA | 25% | Debiased | 0.945 | 0.007 | 0.946 | 0.934 | 0.955 |
| NA | 50% | LLM | 0.865 | 0.099 | 0.913 | 0.653 | 0.947 |
| NA | 50% | Human Validation | 0.949 | 0.007 | 0.948 | 0.939 | 0.959 |
| NA | 50% | Debiased | 0.948 | 0.007 | 0.949 | 0.937 | 0.957 |

Table 44: Coverage Summary Statistics of $\beta_{14}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.685 | 0.225 | 0.764 | 0.252 | 0.932 |
| NA | 5% | Human Validation | 0.917 | 0.009 | 0.917 | 0.901 | 0.930 |
| NA | 5% | Debiased | 0.938 | 0.010 | 0.939 | 0.920 | 0.953 |
| NA | 10% | LLM | 0.689 | 0.226 | 0.770 | 0.242 | 0.937 |
| NA | 10% | Human Validation | 0.933 | 0.008 | 0.934 | 0.918 | 0.945 |
| NA | 10% | Debiased | 0.940 | 0.008 | 0.942 | 0.926 | 0.951 |
| NA | 25% | LLM | 0.686 | 0.225 | 0.759 | 0.237 | 0.938 |
| NA | 25% | Human Validation | 0.943 | 0.008 | 0.944 | 0.929 | 0.955 |
| NA | 25% | Debiased | 0.946 | 0.007 | 0.946 | 0.937 | 0.957 |
| NA | 50% | LLM | 0.690 | 0.225 | 0.766 | 0.255 | 0.935 |
| NA | 50% | Human Validation | 0.947 | 0.007 | 0.948 | 0.936 | 0.959 |
| NA | 50% | Debiased | 0.948 | 0.007 | 0.949 | 0.937 | 0.958 |

Table 45: Coverage Summary Statistics of $\beta_{15}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.925 | 0.032 | 0.936 | 0.879 | 0.957 |
| NA | 5% | Human Validation | 0.932 | 0.009 | 0.931 | 0.919 | 0.946 |
| NA | 5% | Debiased | 0.932 | 0.010 | 0.933 | 0.919 | 0.950 |
| NA | 10% | LLM | 0.926 | 0.030 | 0.938 | 0.876 | 0.953 |
| NA | 10% | Human Validation | 0.940 | 0.008 | 0.941 | 0.927 | 0.951 |
| NA | 10% | Debiased | 0.941 | 0.008 | 0.941 | 0.928 | 0.955 |
| NA | 25% | LLM | 0.925 | 0.033 | 0.938 | 0.868 | 0.953 |
| NA | 25% | Human Validation | 0.946 | 0.007 | 0.946 | 0.934 | 0.958 |
| NA | 25% | Debiased | 0.947 | 0.007 | 0.947 | 0.934 | 0.957 |
| NA | 50% | LLM | 0.926 | 0.031 | 0.933 | 0.883 | 0.956 |
| NA | 50% | Human Validation | 0.947 | 0.006 | 0.947 | 0.938 | 0.958 |
| NA | 50% | Debiased | 0.947 | 0.007 | 0.947 | 0.934 | 0.957 |

Table 46: Coverage Summary Statistics of $\beta_{19}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.891 | 0.065 | 0.915 | 0.740 | 0.954 |
| NA | 5% | Human Validation | 0.937 | 0.009 | 0.938 | 0.926 | 0.953 |
| NA | 5% | Debiased | 0.944 | 0.009 | 0.944 | 0.931 | 0.956 |
| NA | 10% | LLM | 0.890 | 0.066 | 0.912 | 0.750 | 0.949 |
| NA | 10% | Human Validation | 0.943 | 0.008 | 0.943 | 0.930 | 0.954 |
| NA | 10% | Debiased | 0.947 | 0.008 | 0.947 | 0.934 | 0.957 |
| NA | 25% | LLM | 0.894 | 0.064 | 0.915 | 0.737 | 0.954 |
| NA | 25% | Human Validation | 0.947 | 0.007 | 0.947 | 0.938 | 0.958 |
| NA | 25% | Debiased | 0.947 | 0.008 | 0.948 | 0.935 | 0.959 |
| NA | 50% | LLM | 0.889 | 0.066 | 0.913 | 0.735 | 0.947 |
| NA | 50% | Human Validation | 0.951 | 0.007 | 0.950 | 0.940 | 0.961 |
| NA | 50% | Debiased | 0.947 | 0.007 | 0.948 | 0.935 | 0.957 |

Table 47: Coverage Summary Statistics of $\beta_{20}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.901 | 0.062 | 0.923 | 0.784 | 0.956 |
| NA | 5% | Human Validation | 0.938 | 0.007 | 0.938 | 0.926 | 0.950 |
| NA | 5% | Debiased | 0.938 | 0.008 | 0.938 | 0.926 | 0.951 |
| NA | 10% | LLM | 0.902 | 0.056 | 0.920 | 0.785 | 0.953 |
| NA | 10% | Human Validation | 0.943 | 0.007 | 0.942 | 0.932 | 0.956 |
| NA | 10% | Debiased | 0.945 | 0.007 | 0.944 | 0.934 | 0.956 |
| NA | 25% | LLM | 0.903 | 0.058 | 0.925 | 0.783 | 0.952 |
| NA | 25% | Human Validation | 0.948 | 0.007 | 0.948 | 0.937 | 0.959 |
| NA | 25% | Debiased | 0.947 | 0.007 | 0.947 | 0.937 | 0.959 |
| NA | 50% | LLM | 0.904 | 0.055 | 0.923 | 0.796 | 0.954 |
| NA | 50% | Human Validation | 0.950 | 0.007 | 0.951 | 0.938 | 0.960 |
| NA | 50% | Debiased | 0.949 | 0.007 | 0.948 | 0.939 | 0.961 |

Table 48: Coverage Summary Statistics of $\beta_{\text{Other}}$ by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model | Validation Proportion | MSE | Mean | SD | Median | 5% | 95% |
|---|---|---|---|---|---|---|---|
| NA | 5% | LLM | 0.836 | 0.133 | 0.886 | 0.570 | 0.950 |
| NA | 5% | Human Validation | 0.949 | 0.007 | 0.949 | 0.937 | 0.961 |
| NA | 5% | Debiased | 0.954 | 0.007 | 0.954 | 0.943 | 0.966 |
| NA | 10% | LLM | 0.841 | 0.133 | 0.891 | 0.550 | 0.955 |
| NA | 10% | Human Validation | 0.950 | 0.007 | 0.950 | 0.939 | 0.960 |
| NA | 10% | Debiased | 0.951 | 0.008 | 0.952 | 0.938 | 0.962 |
| NA | 25% | LLM | 0.839 | 0.133 | 0.887 | 0.581 | 0.949 |
| NA | 25% | Human Validation | 0.949 | 0.007 | 0.950 | 0.938 | 0.958 |
| NA | 25% | Debiased | 0.948 | 0.007 | 0.950 | 0.935 | 0.959 |
| NA | 50% | LLM | 0.840 | 0.129 | 0.891 | 0.561 | 0.951 |
| NA | 50% | Human Validation | 0.951 | 0.007 | 0.950 | 0.940 | 0.962 |
| NA | 50% | Debiased | 0.949 | 0.007 | 0.949 | 0.939 | 0.960 |